

Expectation

65 problems

36 textbook exercises · 21 strategic practice · 8 homework

Questions below; full solutions at the end of the chapter.

TEXTBOOK EXERCISES

Expectations and variances

Q1 Book Ch.4, Exercise 13

[solution · p. 111](#)

13. (S) Are there discrete random variables X and Y such that $E(X) > 100E(Y)$ but Y is greater than X with probability at least 0.99?

Named distributions

Q2 Book Ch.4, Exercise 17

[solution · p. 111](#)

17. (S) A couple decides to keep having children until they have at least one boy and at least one girl, and then stop. Assume they never have twins, that the “trials” are independent with probability $1/2$ of a boy, and that they are fertile enough to keep producing children indefinitely. What is the expected number of children?

Q3 Book Ch.4, Exercise 18

[solution · p. 111](#)

18. (S) A coin is tossed repeatedly until it lands Heads for the first time. Let X be the number of tosses that are required (including the toss that landed Heads), and let p be the probability of Heads, so that $X \sim \text{FS}(p)$. Find the CDF of X , and for $p = 1/2$ sketch its graph.

20. (S) Let $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n+1, \frac{1}{2})$, independently. (This problem has been revised from that in the first printing of the book, to avoid overlap with Exercise 3.25.)
- (a) Let $V = \min(X, Y)$ be the smaller of X and Y , and let $W = \max(X, Y)$ be the larger of X and Y . So if X crystallizes to x and Y crystallizes to y , then V crystallizes to $\min(x, y)$ and W crystallizes to $\max(x, y)$. Find $E(V) + E(W)$.
- (b) Show that $E|X - Y| = E(W) - E(V)$, with notation as in (a).
- (c) Compute $\text{Var}(n - X)$ in two different ways.

21. (S) Raindrops are falling at an average rate of 20 drops per square inch per minute. What would be a reasonable distribution to use for the number of raindrops hitting a particular region measuring 5 inches² in t minutes? Why? Using your chosen distribution, compute the probability that the region has no rain drops in a given 3-second time interval.

22. (S) Alice and Bob have just met, and wonder whether they have a mutual friend. Each has 50 friends, out of 1000 other people who live in their town. They think that it's unlikely that they have a friend in common, saying "each of us is only friends with 5% of the people here, so it would be very unlikely that our two 5%'s overlap."
- Assume that Alice's 50 friends are a random sample of the 1000 people (equally likely to be any 50 of the 1000), and similarly for Bob. Also assume that knowing who Alice's friends are gives no information about who Bob's friends are.
- (a) Compute the expected number of mutual friends Alice and Bob have.
- (b) Let X be the number of mutual friends they have. Find the PMF of X .
- (c) Is the distribution of X one of the important distributions we have looked at? If so, which?

24. (S) Calvin and Hobbes play a match consisting of a series of games, where Calvin has probability p of winning each game (independently). They play with a "win by two" rule: the first player to win two games more than his opponent wins the match. Find the expected number of games played.
- Hint: Consider the first two games as a pair, then the next two as a pair, etc.

26. (S) Let X and Y be $\text{Pois}(\lambda)$ r.v.s, and $T = X + Y$. Suppose that X and Y are *not* independent, and in fact $X = Y$. Prove or disprove the claim that $T \sim \text{Pois}(2\lambda)$ in this scenario.

29. (S) A discrete distribution has the *memoryless property* if for X a random variable with that distribution, $P(X \geq j + k | X \geq j) = P(X \geq k)$ for all nonnegative integers j, k .
- (a) If X has a memoryless distribution with CDF F and PMF $p_i = P(X = i)$, find an expression for $P(X \geq j + k)$ in terms of $F(j), F(k), p_j, p_k$.
- (b) Name a discrete distribution which has the memoryless property. Justify your answer with a clear interpretation in words or with a computation.

Indicator r.v.s

30. (S) Randomly, k distinguishable balls are placed into n distinguishable boxes, with all possibilities equally likely. Find the expected number of empty boxes.

31. (S) A group of 50 people are comparing their birthdays (as usual, assume their birthdays are independent, are not February 29, etc.). Find the expected number of pairs of people with the same birthday, and the expected number of days in the year on which at least two of these people were born.

32. (S) A group of $n \geq 4$ people are comparing their birthdays (as usual, assume their birthdays are independent, are not February 29, etc.). Let I_{ij} be the indicator r.v. of i and j having the same birthday (for $i < j$). Is I_{12} independent of I_{34} ? Is I_{12} independent of I_{13} ? Are the I_{ij} independent?

33. (S) A total of 20 bags of Haribo gummi bears are randomly distributed to 20 students. Each bag is obtained by a random student, and the outcomes of who gets which bag are independent. Find the average number of bags of gummi bears that the first three students get in total, and find the average number of students who get at least one bag.

40. (S) There are 100 shoelaces in a box. At each stage, you pick two random ends and tie them together. Either this results in a longer shoelace (if the two ends came from different pieces), or it results in a loop (if the two ends came from the same piece). What are the expected number of steps until everything is in loops, and the expected number of loops after everything is in loops? (This is a famous interview problem; leave the latter answer as a sum.)

Hint: For each step, create an indicator r.v. for whether a loop was created then, and note that the number of free ends goes down by 2 after each step.

44. ⑤ Let X be Hypergeometric with parameters w, b, n .
- (a) Find $E\binom{X}{2}$ by thinking, without any complicated calculations.
- (b) Use (a) to find the variance of X . You should get

$$\text{Var}(X) = \frac{N-n}{N-1}npq,$$

where $N = w + b, p = w/N, q = 1 - p$.

Q16 Book Ch.4, Exercise 47

solution · p. 115

47. ⑤ A hash table is being used to store the phone numbers of k people, storing each person's phone number in a uniformly random location, represented by an integer between 1 and n (see Exercise 25 from Chapter 1 for a description of hash tables). Find the expected number of locations with no phone numbers stored, the expected number with exactly one phone number, and the expected number with more than one phone number (should these quantities add up to n ?).

Q17 Book Ch.4, Exercise 50

solution · p. 115

50. ⑤ Consider the following algorithm, known as *bubble sort*, for sorting a list of n distinct numbers into increasing order. Initially they are in a random order, with all orders equally likely. The algorithm compares the numbers in positions 1 and 2, and swaps them if needed, then it compares the new numbers in positions 2 and 3, and swaps them if needed, etc., until it has gone through the whole list. Call this one “sweep” through the list. After the first sweep, the largest number is at the end, so the second sweep (if needed) only needs to work with the first $n - 1$ positions. Similarly, the third sweep (if needed) only needs to work with the first $n - 2$ positions, etc. Sweeps are performed until $n - 1$ sweeps have been completed or there is a swapless sweep.

For example, if the initial list is 53241 (omitting commas), then the following 4 sweeps are performed to sort the list, with a total of 10 comparisons:

53241 $\rightarrow$ 35241 $\rightarrow$ 32541 $\rightarrow$ 32451 $\rightarrow$ 32415.

32415 $\rightarrow$ 23415 $\rightarrow$ 23415 $\rightarrow$ 23145.

23145 $\rightarrow$ 23145 $\rightarrow$ 21345.

21345 $\rightarrow$ 12345.

(a) An *inversion* is a pair of numbers that are out of order (e.g., 12345 has no inversions, while 53241 has 8 inversions). Find the expected number of inversions in the original list.

(b) Show that the expected number of comparisons is between $\frac{1}{2}\binom{n}{2}$ and $\binom{n}{2}$.

Hint: For one bound, think about how many comparisons are made if $n - 1$ sweeps are done; for the other bound, use Part (a).

Q18 Book Ch.4, Exercise 52

solution · p. 116

52. ⑤ An urn contains red, green, and blue balls. Balls are chosen randomly with replacement (each time, the color is noted and then the ball is put back). Let r, g, b be the probabilities of drawing a red, green, blue ball, respectively ($r + g + b = 1$).
- (a) Find the expected number of balls chosen before obtaining the first red ball, not including the red ball itself.
- (b) Find the expected number of different *colors* of balls obtained before getting the first red ball.
- (c) Find the probability that at least 2 of n balls drawn are red, given that at least 1 is red.

53. ⑤ Job candidates $C_1, C_2, \dots$ are interviewed one by one, and the interviewer compares them and keeps an updated list of rankings (if n candidates have been interviewed so far, this is a list of the n candidates, from best to worst). Assume that there is no limit on the number of candidates available, that for any n the candidates $C_1, C_2, \dots, C_n$ are equally likely to arrive in any order, and that there are no ties in the rankings given by the interview.

Let X be the index of the first candidate to come along who ranks as better than the very first candidate C_1 (so C_X is better than C_1 , but the candidates after 1 but prior to X (if any) are worse than C_1). For example, if C_2 and C_3 are worse than C_1 but C_4 is better than C_1 , then $X = 4$. All $4!$ orderings of the first 4 candidates are equally likely, so it could have happened that the first candidate was the best out of the first 4 candidates, in which case $X > 4$.

What is $E(X)$ (which is a measure of how long, on average, the interviewer needs to wait to find someone better than the very first candidate)?

Hint: Find $P(X > n)$ by interpreting what $X > n$ says about how C_1 compares with other candidates, and then apply the result of Theorem 4.4.8.

LOTUS

56. ⑤ For $X \sim \text{Pois}(\lambda)$, find $E(X!)$ (the average factorial of X), if it is finite.

59. ⑤ Let $X \sim \text{Geom}(p)$ and let t be a constant. Find $E(e^{tX})$, as a function of t (this is known as the *moment generating function*; we will see in Chapter 6 how this function is useful).

60. ⑤ The number of fish in a certain lake is a $\text{Pois}(\lambda)$ random variable. Worried that there might be no fish at all, a statistician adds one fish to the lake. Let Y be the resulting number of fish (so Y is 1 plus a $\text{Pois}(\lambda)$ random variable).
- (a) Find $E(Y^2)$.
- (b) Find $E(1/Y)$.

61. ⑤ Let X be a $\text{Pois}(\lambda)$ random variable, where λ is fixed but unknown. Let $\theta = e^{-3\lambda}$, and suppose that we are interested in estimating θ based on the data. Since X is what we observe, our estimator is a function of X , call it $g(X)$. The *bias* of the estimator $g(X)$ is defined to be $E(g(X)) - \theta$, i.e., how far off the estimate is on average; the estimator is *unbiased* if its bias is 0.
- (a) For estimating λ , the r.v. X itself is an unbiased estimator. Compute the bias of the estimator $T = e^{-3X}$. Is it unbiased for estimating θ ?
- (b) Show that $g(X) = (-2)^X$ is an unbiased estimator for θ . (In fact, it turns out to be the only unbiased estimator for θ .)
- (c) Explain intuitively why $g(X)$ is a silly choice for estimating θ , despite (b), and show how to improve it by finding an estimator $h(X)$ for θ that is always at least as good as $g(X)$ and sometimes strictly better than $g(X)$. That is,

$$|h(X) - \theta| \leq |g(X) - \theta|,$$

with the inequality sometimes strict.

Poisson approximation

Q24 Book Ch.4, Exercise 62

[solution · p. 117](#)

62. ⑤ Law school courses often have assigned seating to facilitate the Socratic method. Suppose that there are 100 first-year law students, and each takes the same two courses: Torts and Contracts. Both are held in the same lecture hall (which has 100 seats), and the seating is uniformly random and independent for the two courses.
- (a) Find the probability that no one has the same seat for both courses (exactly; you should leave your answer as a sum).
- (b) Find a simple but accurate approximation to the probability that no one has the same seat for both courses.
- (c) Find a simple but accurate approximation to the probability that at least two students have the same seat for both courses.

Q25 Book Ch.4, Exercise 63

[solution · p. 118](#)

63. ⑤ A group of n people play “Secret Santa” as follows: each puts his or her name on a slip of paper in a hat, picks a name randomly from the hat (without replacement), and then buys a gift for that person. Unfortunately, they overlook the possibility of drawing one’s own name, so some may have to buy gifts for themselves (on the bright side, some may like self-selected gifts better). Assume $n \geq 2$.
- (a) Find the expected value of the number X of people who pick their own names.
- (b) Find the expected number of pairs of people, A and B , such that A picks B ’s name and B picks A ’s name (where $A \neq B$ and order doesn’t matter).
- (c) Let X be the number of people who pick their own names. What is the *approximate* distribution of X if n is large (specify the parameter value or values)? What does $P(X = 0)$ converge to as $n \rightarrow \infty$?

65. (S) Ten million people enter a certain lottery. For each person, the chance of winning is one in ten million, independently.
- (a) Find a simple, good approximation for the PMF of the number of people who win the lottery.
- (b) Congratulations! You won the lottery. However, there may be other winners. Assume now that the number of winners other than you is $W \sim \text{Pois}(1)$, and that if there is more than one winner, then the prize is awarded to one randomly chosen winner. Given this information, find the probability that you win the prize (simplify).

***Existence**

68. (S) Each of 111 people names his or her 5 favorite movies out of a list of 11 movies.
- (a) Alice and Bob are 2 of the 111 people. Assume *for this part only* that Alice's 5 favorite movies out of the 11 are random, with all sets of 5 equally likely, and likewise for Bob, independently. Find the expected number of movies in common to Alice's and Bob's lists of favorite movies.
- (b) Show that there are 2 movies such that at least 21 of the people name both of these movies as favorites.

69. (S) The circumference of a circle is colored with red and blue ink such that $2/3$ of the circumference is red and $1/3$ is blue. Prove that no matter how complicated the coloring scheme is, there is a way to inscribe a square in the circle such that at least three of the four corners of the square touch red ink.

70. (S) A hundred students have taken an exam consisting of 8 problems, and for each problem at least 65 of the students got the right answer. Show that there exist two students who collectively got everything right, in the sense that for each problem, at least one of the two got it right.

71. (S) Ten points in the plane are designated. You have ten circular coins (of the same radius). Show that you can position the coins in the plane (without stacking them) so that all ten points are covered.

Hint: Consider a *honeycomb tiling* of the plane (this is a way to divide the plane into hexagons). You can use the fact from geometry that if a circle is inscribed in a hexagon then the ratio of the area of the circle to the area of the hexagon is $\frac{\pi}{2\sqrt{3}} > 0.9$.

72. ⑤ Let S be a set of binary strings $a_1 \dots a_n$ of length n (where juxtaposition means concatenation). We call S k -complete if for any indices $1 \leq i_1 < \dots < i_k \leq n$ and any binary string $b_1 \dots b_k$ of length k , there is a string $s_1 \dots s_n$ in S such that $s_{i_1} s_{i_2} \dots s_{i_k} = b_1 b_2 \dots b_k$. For example, for $n = 3$, the set $S = \{001, 010, 011, 100, 101, 110\}$ is 2-complete since all 4 patterns of 0's and 1's of length 2 can be found in any 2 positions. Show that if $\binom{n}{k} 2^k (1 - 2^{-k})^m < 1$, then there exists a k -complete set of size at most m .

Mixed practice

Q32 Book Ch.4, Exercise 75

[solution · p. 120](#)

75. ⑤ A group of 360 people is going to be split into 120 teams of 3 (where the order of teams and the order within a team don't matter).
- (a) How many ways are there to do this?
- (b) The group consists of 180 married couples. A random split into teams of 3 is chosen, with all possible splits equally likely. Find the expected number of teams containing married couples.

Q33 Book Ch.4, Exercise 76

[solution · p. 121](#)

76. ⑤ The gambler de Méré asked Pascal whether it is more likely to get at least one six in 4 rolls of a die, or to get at least one double-six in 24 rolls of a pair of dice. Continuing this pattern, suppose that a group of n fair dice is rolled $4 \cdot 6^{n-1}$ times.
- (a) Find the expected number of times that “all sixes” is achieved (i.e., how often among the $4 \cdot 6^{n-1}$ rolls it happens that all n dice land 6 simultaneously).
- (b) Give a simple but accurate approximation of the probability of having at least one occurrence of “all sixes”, for n large (in terms of e but not n).
- (c) de Méré finds it tedious to re-roll so many dice. So after one normal roll of the n dice, in going from one roll to the next, with probability $6/7$ he leaves the dice in the same configuration and with probability $1/7$ he re-rolls. For example, if $n = 3$ and the 7th roll is $(3, 1, 4)$, then $6/7$ of the time the 8th roll remains $(3, 1, 4)$ and $1/7$ of the time the 8th roll is a new random outcome. Does the expected number of times that “all sixes” is achieved stay the same, increase, or decrease (compared with (a))? Give a short but clear explanation.

Q34 Book Ch.4, Exercise 77

[solution · p. 121](#)

77. ⑤ Five people have just won a \$100 prize, and are deciding how to divide the \$100 up between them. Assume that whole dollars are used, not cents. Also, for example, giving \$50 to the first person and \$10 to the second is different from vice versa.
- (a) How many ways are there to divide up the \$100, such that each gets at least \$10?
- (b) Assume that the \$100 is randomly divided up, with all of the possible allocations counted in (a) equally likely. Find the expected amount of money that the first person receives.
- (c) Let A_j be the event that the j th person receives more than the first person (for $2 \leq j \leq 5$), when the \$100 is randomly allocated as in (b). Are A_2 and A_3 independent?

78. ⑤ Joe's iPod has 500 different songs, consisting of 50 albums of 10 songs each. He listens to 11 random songs on his iPod, with all songs equally likely and chosen independently (so repetitions may occur).
- What is the PMF of how many of the 11 songs are from his favorite album?
 - What is the probability that there are 2 (or more) songs from the same album among the 11 songs he listens to?
 - A pair of songs is a *match* if they are from the same album. If, say, the 1st, 3rd, and 7th songs are all from the same album, this counts as 3 matches. Among the 11 songs he listens to, how many matches are there on average?

79. ⑤ In each day that the Mass Cash lottery is run in Massachusetts, 5 of the integers from 1 to 35 are chosen (randomly and without replacement).
- When playing this lottery, find the probability of guessing exactly 3 numbers right, given that you guess at least 1 of the numbers right.
 - Find an exact expression for the expected number of days needed so that all of the $\binom{35}{5}$ possible lottery outcomes will have occurred.
 - Approximate the probability that after 50 days of the lottery, every number from 1 to 35 has been picked at least once.

STRATEGIC PRACTICE

1 Distributions and Expected Values for Discrete Random Variables

- Find an example of two discrete random variables X and Y (on the same sample space) such that X and Y have the same distribution (i.e., same PMF and same CDF), but the event $X = Y$ *never* occurs.

- Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, etc. (so X takes values $1, 2, \dots, 7$, with equal probabilities). Let Y be the next day after X (again represented as an integer between 1 and 7). Do X and Y have the same distribution? What is $P(X < Y)$?

- A coin is tossed repeatedly until it lands Heads for the first time. Let X be the number of tosses that are required (including the toss that landed Heads), and let p be the probability of Heads. Find the CDF of X , and for $p = 1/2$ sketch its graph.

4. Are there discrete random variables X and Y such that $E(X) > 100E(Y)$ but Y is greater than X with probability at least 0.99?

5. Let X be a discrete r.v. with possible values $1, 2, 3, \dots$. Let $F(x) = P(X \leq x)$ be the CDF of X . Show that

$$E(X) = \sum_{n=0}^{\infty} (1 - F(n)).$$

Hint: organize the order of summation carefully, using the fact that, for example, $P(X > 3) = P(X = 4) + P(X = 5) + \dots$.

6. Job candidates $C_1, C_2, \dots$ are interviewed one by one, and the interviewer compares them and keeps an updated list of rankings (if n candidates have been interviewed so far, this is a list of the n candidates, from best to worst). Assume that there is no limit on the number of candidates available, that for any n the candidates $C_1, C_2, \dots, C_n$ are equally likely to arrive in any order, and that there are no ties in the rankings given by the interview.

Let X be the index of the first candidate to come along who ranks as better than the very first candidate C_1 (so C_X is better than C_1 , but the candidates after 1 but prior to X (if any) are worse than C_1). For example, if C_2 and C_3 are worse than C_1 but C_4 is better than C_1 , then $X = 4$. All $4!$ orderings of the first 4 candidates are equally likely, so it could have happened that the first candidate was the best out of the first 4 candidates, in which case $X > 4$.

What is $E(X)$ (which is a measure of how long, on average, the interviewer needs to wait to find someone better than the very first candidate)? Hint: find $P(X > n)$ by interpreting what $X > n$ says about how C_1 compares with other candidates, and then apply the result of the previous problem.

2 Indicator Random Variables and Linearity of Expectation

1. A group of 50 people are comparing their birthdays (as usual, assume their birthdays are independent, are not February 29, etc.). Find the expected number of pairs of people with the same birthday, and the expected number of days in the year on which at least two of these people were born.

2. A total of 20 bags of Haribo gummi bears are randomly distributed to the 20 students in a certain Stat 110 section. Each bag is obtained by a random student, and the outcomes of who gets which bag are independent. Find the average number of bags of gummi bears that the first three students get in total, and find the average number of students who get at least one bag.

3. There are 100 shoelaces in a box. At each stage, you pick two random ends and tie them together. Either this results in a longer shoelace (if the two ends came from different pieces), or it results in a loop (if the two ends came from the same piece). What are the expected number of steps until everything is in loops, and the expected number of loops after everything is in loops? (This is a famous interview problem; leave the latter answer as a sum.)

Hint: for each step, create an indicator r.v. for whether a loop was created then, and note that the number of free ends goes down by 2 after each step.

4. A *hash table* is a commonly used data structure in computer science, allowing for fast information retrieval. For example, suppose we want to store some people's phone numbers. Assume that no two of the people have the same name. For each name x , a *hash function* h is used, where $h(x)$ is the location to store x 's phone number. After such a table has been computed, to look up x 's phone number one just recomputes $h(x)$ and then looks up what is stored in that location.

The hash function h is deterministic, since we don't want to get different results every time we compute $h(x)$. But h is often chosen to be *pseudorandom*. For this problem, assume that true randomness is used. So let there be k people, with each person's phone number stored in a random location (independently), represented by an integer between 1 and n . It then might happen that one location has more than one phone number stored there, if two different people x and y end up with the same random location for their information to be stored.

Find the expected number of locations with no phone numbers stored, the expected number with exactly one phone number, and the expected number with more than one phone number (should these quantities add up to n ?).

1. Raindrops are falling at an average rate of 20 drops per square inch per minute. What would be a reasonable distribution to use for the number of raindrops hitting a particular region measuring 5 inches² in t minutes? Why? Using your chosen distribution, compute the probability that the region has no rain drops in a given 3 second time interval.

2. Harvard Law School courses often have assigned seating to facilitate the “Socratic method.” Suppose that there are 100 first year Harvard Law students, and each takes two courses: Torts and Contracts. Both are held in the same lecture hall (which has 100 seats), and the seating is uniformly random and independent for the two courses.
 - (a) Find the probability that no one has the same seat for both courses (exactly; you should leave your answer as a sum).
 - (b) Find a simple but accurate approximation to the probability that no one has the same seat for both courses.
 - (c) Find a simple but accurate approximation to the probability that at least two students have the same seat for both courses.

3. Let X be a $\text{Pois}(\lambda)$ random variable, where λ is fixed but unknown. Let $\theta = e^{-3\lambda}$, and suppose that we are interested in estimating θ based on the data. Since X is what we observe, our estimator is a function of X , call it $g(X)$. The *bias* of the estimator $g(X)$ is defined to be $E(g(X)) - \theta$, i.e., how far off the estimate is on average; the estimator is *unbiased* if its bias is 0.
 - (a) For estimating λ , the r.v. X itself is an unbiased estimator. Compute the bias of the estimator $T = e^{-3X}$. Is it unbiased for estimating θ ?
 - (b) Show that $g(X) = (-2)^X$ is an unbiased estimator for θ . (In fact, it is the best unbiased estimator, in the sense of minimizing the average squared error.)
 - (c) Explain intuitively why $g(X)$ is a silly choice for estimating θ , despite (b), and show how to improve it by finding an estimator $h(X)$ for θ that is always at least as good as $g(X)$ and sometimes strictly better than $g(X)$. That is,

$$|h(X) - \theta| \leq |g(X) - \theta|,$$

with the inequality sometimes strict.

2 Seeking Sublime Symmetry

Q50 SP 5 §2 · Problem 1

[solution · p. 131](#)

1. Let $Z \sim \mathcal{N}(0, 1)$ and let S be a “random sign” independent of Z , i.e., S is 1 with probability $1/2$ and -1 with probability $1/2$. Show that $SZ \sim \mathcal{N}(0, 1)$.

Q51 SP 5 §2 · Problem 2

[solution · p. 132](#)

2. Explain why $P(X < Y) = P(Y < X)$ if X and Y are i.i.d. Does it follow that $P(X < Y) = 1/2$? Is it still always true that $P(X < Y) = P(Y < X)$ if X and Y have the same distribution but are not independent?

Q52 SP 5 §2 · Problem 3

[solution · p. 132](#)

3. Explain why if $X \sim \text{Bin}(n, p)$, then $n - X \sim \text{Bin}(n, 1 - p)$.

Q53 SP 5 §2 · Problem 4

[solution · p. 132](#)

4. There are 100 passengers lined up to board an airplane with 100 seats (with each seat assigned to one of the passengers). The first passenger in line crazily decides to sit in a randomly chosen seat (with all seats equally likely). Each subsequent passenger takes his or her assigned seat if available, and otherwise sits in a random available seat. What is the probability that the last passenger in line gets to sit in his or her assigned seat? (This is another common interview problem, and a beautiful example of the power of symmetry.)

Hint: call the seat assigned to the j th passenger in line “Seat j ” (regardless of whether the airline calls it seat 23A or whatever). What are the possibilities for which seats are available to the last passenger in line, and what is the probability of each of these possibilities?

3 Existence

Q54 SP 8 §3 · Problem 1

[solution · p. 133](#)

1. Let S be a set of binary strings $a_1 \dots a_n$ of length n (where juxtaposition means concatenation). We call S k -complete if for any indices $1 \leq i_1 < \dots < i_k \leq n$ and any binary string $b_1 \dots b_k$ of length k , there is a string $s_1 \dots s_n$ in S such that $s_{i_1} s_{i_2} \dots s_{i_k} = b_1 b_2 \dots b_k$. For example, for $n = 3$, the set $S = \{001, 010, 011, 100, 101, 110\}$ is 2-complete since all 4 patterns of 0’s and 1’s of length 2 can be found in any 2 positions. Show that if $\binom{n}{k} 2^k (1 - 2^{-k})^m < 1$, then there exists a k -complete set of size at most m .

2. A hundred students have taken an exam consisting of 8 problems, and for each problem at least 65 of the students got the right answer. Show that there exist two students who collectively got everything right, in the sense that for each problem, at least one of the two got it right.

3. The circumference of a circle is colored with red and blue ink such that $2/3$ of the circumference is red and $1/3$ is blue. Prove that no matter how complicated the coloring scheme is, there is a way to inscribe a square in the circle such that at least three of the four corners of the square touch red ink.

4. Ten points in the plane are designated. You have ten circular coins (of the same radius). Show that you can position the coins in the plane (without stacking them) so that all ten points are covered.

Hint: consider a honeycomb tiling as in <http://mathworld.wolfram.com/Honeycomb>. You can use the fact from geometry that if a circle is inscribed in a hexagon then the ratio of the area of the circle to the area of the hexagon is $\frac{\pi}{2\sqrt{3}} > 0.9$.

HOMEWORK

3. A couple decides to keep having children until they have at least one boy and at least one girl, and then stop. Assume they never have twins, that the “trials” are independent with probability $1/2$ of a boy, and that they are fertile enough to keep producing children indefinitely. What is the expected number of children?

4. Randomly, k distinguishable balls are placed into n distinguishable boxes, with all possibilities equally likely. Find the expected number of empty boxes.

6. Consider the following algorithm for sorting a list of n distinct numbers into increasing order. Initially they are in a random order, with all orders equally likely. The algorithm compares the numbers in positions 1 and 2, and swaps them if needed, then it compares the new numbers in positions 2 and 3, and swaps them if needed, etc., until it has gone through the whole list. Call this one “sweep” through the list. After the first sweep, the largest number is at the end, so the second sweep (if needed) only needs to work with the first $n - 1$ positions. Similarly, the third sweep (if needed) only needs to work with the first $n - 2$ positions, etc. Sweeps are performed until $n - 1$ sweeps have been completed or there is a swapless sweep.

For example, if the initial list is 53241 (omitting commas), then the following 4 sweeps are performed to sort the list, with a total of 10 comparisons:

$$53241 \rightarrow 35241 \rightarrow 32541 \rightarrow 32451 \rightarrow 32415.$$

$$32415 \rightarrow 23415 \rightarrow 23415 \rightarrow 23145.$$

$$23145 \rightarrow 23145 \rightarrow 21345.$$

$$21345 \rightarrow 12345.$$

(a) An *inversion* is a pair of numbers that are out of order (e.g., 12345 has no inversions, while 53241 has 8 inversions). Find the expected number of inversions in the original list.

(b) Show that the expected number of comparisons is between $\frac{1}{2}\binom{n}{2}$ and $\binom{n}{2}$.

Hint for (b): for one bound, think about how many comparisons are made if $n - 1$ sweeps are done; for the other bound, use Part (a).

Q61 HW 4 · Problem 7

[solution · p. 137](#)

7. Athletes compete one at a time at the high jump. Let X_j be how high the j th jumper jumped, with $X_1, X_2, \dots$ i.i.d. with a continuous distribution. We say that the j th jumper set a *record* if X_j is greater than all of $X_{j-1}, \dots, X_1$.

(a) Is the event “the 110th jumper sets a record” independent of the event “the 111th jumper sets a record”? Justify your answer by finding the relevant probabilities in the definition of independence *and* with an intuitive explanation.

(b) Find the mean number of records among the first n jumpers (as a sum). What happens to the mean as $n \rightarrow \infty$?

Q62 HW 5 · Problem 1

[solution · p. 137](#)

1. A group of $n \geq 4$ people are comparing their birthdays (as usual, assume their birthdays are independent, are not February 29, etc.).

(a) Let I_{ij} be the indicator r.v. of i and j having the same birthday (for $i < j$). Is I_{12} independent of I_{34} ? Is I_{12} independent of I_{13} ? Are the I_{ij} 's independent?

(b) Explain why the Poisson Paradigm is applicable here even for moderate n , and use it to get a good approximation to the probability of at least 1 match when $n = 23$.

(c) About how many people are needed so that there is a 50% chance (or better) that two either have the same birthday or are only 1 day apart? (Note that this is much harder than the birthday problem to do exactly, but the Poisson Paradigm makes it possible to get fairly accurate approximations quickly.)

Q63 HW 6 · Problem 3

solution · p. 139

3. Consider an experiment where we observe the value of a random variable X , and estimate the value of an unknown constant θ using some random variable $T = g(X)$ that is a function of X . The r.v. T is called an *estimator*. Think of X as the data observed in the experiment, and θ as an unknown parameter related to the distribution of X .

For example, consider the experiment of flipping a coin n times, where the coin has an unknown probability θ of Heads. After the experiment is performed, we have observed the value of $X \sim \text{Bin}(n, \theta)$. The most natural estimator for θ is then X/n .

(a) The *bias* of an estimator T for θ is defined as $b(T) = E(T) - \theta$. The *mean squared error* is the average squared error when using $T(X)$ to estimate θ :

$$\text{MSE}(T) = E(T - \theta)^2.$$

Show that

$$\text{MSE}(T) = \text{Var}(T) + (b(T))^2.$$

This implies that for fixed MSE, lower bias can only be attained at the cost of higher variance and vice versa; this is a form of the *bias-variance tradeoff*, a phenomenon which arises throughout statistics.

(b) Show without using calculus that the constant c that minimizes $E(X - c)^2$ is the expected value of X . (This means that in choosing a single number to summarize X , the mean is the best choice if the goal is to minimize the average squared error.)

(c) For the case that X is continuous with PDF $f(x)$ which is positive everywhere, show that the value of c that minimizes $E|X - c|$ is the median of X (which is the value m with $P(X \leq m) = 1/2$).

Hint: this can be done either with or without calculus. For the calculus method, use LOTUS to write $E|X - c|$ as an integral, and then split the integral into 2 pieces to get rid of the absolute values. Then use the fundamental theorem of calculus (after writing, for example, $\int_{-\infty}^c (c - x)f(x)dx = c \int_{-\infty}^c f(x)dx - \int_{-\infty}^c xf(x)dx$).

7. Shakespeare wrote a total of 884647 words in his known works. Of course, many words are used more than once, and the number of distinct words in Shakespeare's known writings is 31534 (according to one computation). This puts a lower bound on the size of Shakespeare's vocabulary, but it is likely that Shakespeare knew words which he did not use in these known writings.

More specifically, suppose that a new poem of Shakespeare were uncovered, and consider the following (seemingly impossible) problem: give a good prediction of the number of words in the new poem that do not appear anywhere in Shakespeare's previously known works.

The statisticians Ronald Thisted and Bradley Efron studied this problem in a paper called "Did Shakespeare write a newly-discovered poem?", which performed statistical tests to try to determine whether Shakespeare was the author of a poem discovered by a Shakespearean scholar in 1985. A simplified version of their method is developed in the problem below. The method was originally invented by Alan Turing (the founder of computer science) and I.J. Good as part of the effort to break the German Enigma code during World War II.

Let N be the number of distinct words that Shakespeare knew, and assume these words are numbered from 1 to N . Suppose for simplicity that Shakespeare wrote only two plays, A and B . The plays are reasonably long and they are of the same length. Let X_j be the number of times that word j appears in play A , and Y_j be the number of times it appears in play B , for $1 \leq j \leq N$.

(a) Explain why it is reasonable to model X_j as being Poisson, and Y_j as being Poisson with the same parameter as X_j .

(b) Let the numbers of occurrences of the word “eyeball” (which was coined by Shakespeare) in the two plays be independent $\text{Pois}(\lambda)$ r.v.s. Show that the probability that “eyeball” is used in play B but not in play A is

$$e^{-\lambda}(\lambda - \lambda^2/2! + \lambda^3/3! - \lambda^4/4! + \dots).$$

(c) Now assume that λ from (b) is unknown and is itself taken to be a random variable to reflect this uncertainty. So let λ have a PDF f_0 . Let X be the number of times the word “eyeball” appears in play A and Y be the corresponding value for play B . Assume that the conditional distribution of X, Y given λ is that they are independent $\text{Pois}(\lambda)$ r.v.s. Show that the probability that “eyeball” is used in play B but not in play A is the alternating series

$$P(X = 1) - P(X = 2) + P(X = 3) - P(X = 4) + \dots$$

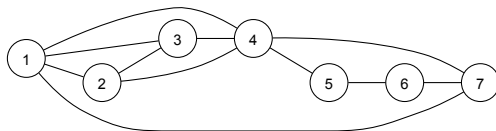
Hint: condition on λ and use (b).

(d) Assume that every word’s numbers of occurrences in A and B are distributed as in (c), where λ may be different for different words but f_0 is fixed. Let W_j be the number of words that appear exactly j times in play A . Show that the expected number of distinct words appearing in play B but not in play A is

$$E(W_1) - E(W_2) + E(W_3) - E(W_4) + \dots$$

(This shows that $W_1 - W_2 + W_3 - W_4 + \dots$ is an *unbiased* predictor of the number of distinct words appearing in play B but not in play A : on average it is correct. Moreover, it can be computed just from having seen play A , without needing to know f_0 or any of the λ_j . This method can be extended in various ways to give predictions for unobserved plays based on observed plays.)

7. A *network* consists of n nodes, each pair of which may or may not have an *edge* joining them. For example, a social network can be modeled as a group of n nodes (representing people), where an edge between i and j means they know each other. Assume the network is undirected and does not have edges from a node to itself (for a social network, this says that if i knows j , then j knows i and that, contrary to Socrates' advice, a person does not know himself or herself). A *clique* of size k is a set of k nodes where every node has an edge to every other node (i.e., within the clique, everyone knows everyone). An *anticlique* of size k is a set of k nodes where there are no edges between them (i.e., within the anticlique, no one knows anyone else). For example, the picture below shows a network with nodes labeled $1, 2, \dots, 7$, where $\{1, 2, 3, 4\}$ is a clique of size 4, and $\{3, 5, 7\}$ is an anticlique of size 3.



(a) Form a random network with n nodes by independently flipping fair coins to decide for each pair $\{x, y\}$ whether there is an edge joining them. Find the expected number of cliques of size k (in terms of n and k).

(b) A *triangle* is a clique of size 3. For a random network as in (a), find the variance of the number of triangles (in terms of n).

Hint: find the covariances of the indicator random variables for each possible clique. There are $\binom{n}{3}$ such indicator r.v.s, some pairs of which are dependent.

(c) Suppose that $\binom{n}{k} < 2^{\binom{k}{2}-1}$. Show that there is a network with n nodes containing no cliques of size k or anticliques of size k .

Hint: explain why it is enough to show that for a random network with n nodes, the probability of the desired property is positive; then consider the complement.

CHAPTER 4 · SOLUTIONS

S1 Book Ch.4, Exercise 13

question · p. 92

Solution: Yes. Consider what happens if we make X usually 0 but on rare occasions, X is extremely large (like the outcome of a lottery); Y , on the other hand, can be more moderate. For a simple example, let X be 10^6 with probability $1/100$ and 0 with probability $99/100$, and let Y be the constant 1 (which is a degenerate r.v.).

S2 Book Ch.4, Exercise 17

question · p. 92

Solution: Let X be the number of children needed, starting with the 2nd child, to obtain one whose gender is not the same as that of the firstborn. Then $X - 1$ is $\text{Geom}(1/2)$, so $E(X) = 2$. This does not include the firstborn, so the expected total number of children is $E(X + 1) = E(X) + 1 = 3$.

Sanity check: An answer of 2 or lower would be a miracle since the couple always needs to have at least 2 children, and sometimes they need more. An answer of 4 or higher would be a miracle since 4 is the expected number of children needed such that there is a boy and a girl with the boy older than the girl.

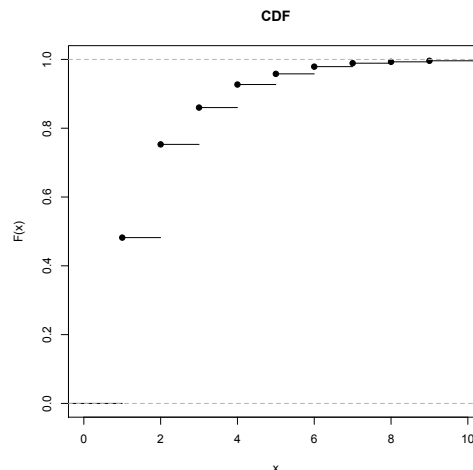
S3 Book Ch.4, Exercise 18

question · p. 92

Solution: By the story of the Geometric, we have $X - 1 \sim \text{Geometric}(p)$. Using this or directly, the PMF is $P(X = k) = p(1 - p)^{k-1}$ for $k \in \{1, 2, 3, \dots\}$ (and 0 otherwise). The CDF can be obtained by adding up the PMF (from $k = 1$ to $k = \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x). We can also see directly that

$$P(X \leq x) = 1 - P(X > x) = 1 - (1 - p)^{\lfloor x \rfloor}$$

for $x \geq 1$, since $X > x$ says that the first $\lfloor x \rfloor$ flips land tails. The CDF is 0 for $x < 1$. For a fair coin, the CDF is $F(x) = 1 - \frac{1}{2^{\lfloor x \rfloor}}$ for $x \geq 1$, and $F(x) = 0$ for $x < 1$, as illustrated below.



Solution:

(a) Note that $V + W = X + Y$ (since adding the smaller and the larger of two numbers is the same as adding both numbers). So by linearity,

$$E(V) + E(W) = E(V + W) = E(X + Y) = E(X) + E(Y) = (2n + 1)/2 = n + \frac{1}{2}.$$

(b) Note that $|X - Y| = W - V$ (since the absolute difference between two numbers is the larger number minus the smaller number). So

$$E|X - Y| = E(W - V) = E(W) - E(V).$$

(c) We have $n - X \sim \text{Bin}(n, 1/2)$, so $\text{Var}(n - X) = n/4$. Alternatively, by properties of variance we have $\text{Var}(n - X) = \text{Var}(n) + \text{Var}(-X) = \text{Var}(X) = n/4$.

S5 Book Ch.4, Exercise 21

question · p. 93

Solution: A reasonable choice of distribution is $\text{Pois}(\lambda t)$, where $\lambda = 20 \cdot 5 = 100$ (the average number of raindrops per minute hitting the region). Assuming this distribution,

$$P(\text{no raindrops in } 1/20 \text{ of a minute}) = e^{-100/20} (100/20)^0 / 0! = e^{-5} \approx 0.0067.$$

S6 Book Ch.4, Exercise 22

question · p. 93

Solution:

(a) Let I_j be the indicator r.v. for the j th person being a mutual friend. Then

$$E\left(\sum_{j=1}^{1000} I_j\right) = 1000E(I_1) = 1000P(I_1 = 1) = 1000 \cdot \left(\frac{5}{100}\right)^2 = 2.5.$$

(b) Condition on who Alice's friends are, and then count the number of ways that Bob can be friends with exactly k of them. This gives

$$P(X = k) = \frac{\binom{50}{k} \binom{950}{50-k}}{\binom{1000}{50}}$$

for $0 \leq k \leq 50$ (and 0 otherwise).

(c) Yes, it is the Hypergeometric distribution, as shown by the PMF from (b) or by thinking of “tagging” Alice's friends (like the elk) and then seeing how many tagged people there are among Bob's friends.

S7 Book Ch.4, Exercise 24

question · p. 93

Solution: Think of the first 2 games, the 3rd and 4th, the 5th and 6th, etc. as “mini-matches.” The match ends right after the first mini-match which isn't a tie. The probability of a mini-match not being a tie is $p^2 + q^2$, so the number of mini-matches needed is 1 plus a $\text{Geom}(p^2 + q^2)$ r.v. Thus, the expected number of games is $\frac{2}{p^2 + q^2}$.

Sanity check: For $p = 0$ or $p = 1$, this reduces to 2. The expected number of games is maximized when $p = \frac{1}{2}$, which makes sense intuitively. Also, it makes sense that the result is symmetric in p and q .

Solution: The r.v. $T = 2X$ is *not* Poisson: it can only take even values $0, 2, 4, 6, \dots$, whereas any Poisson r.v. has positive probability of being any of $0, 1, 2, 3, \dots$.

Alternatively, we can compute the PMF of $2X$, or note that $\text{Var}(2X) = 4\lambda \neq 2\lambda = E(2X)$, whereas for any Poisson r.v. the variance equals the mean.

Solution:

(a) By the memoryless property,

$$P(X \geq k) = P(X \geq j+k | X \geq j) = \frac{P(X \geq j+k, X \geq j)}{P(X \geq j)} = \frac{P(X \geq j+k)}{P(X \geq j)},$$

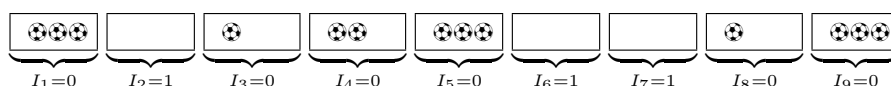
so

$$P(X \geq j+k) = P(X \geq j)P(X \geq k) = (1 - F(j) + p_j)(1 - F(k) + p_k).$$

(b) The Geometric distribution is memoryless (in fact, it turns out to be essentially the *only* discrete memoryless distribution!). This follows from the story of the Geometric: consider Bernoulli trials, waiting for the first success (and defining waiting time to be the number of failures before the first success). Say we have already had j failures without a success. Then the additional waiting time from that point forward has the same distribution as the original waiting time (the Bernoulli trials neither are conspiring against the experimenter nor act as if he or she is due for a success: the trials are independent). A calculation agrees: for $X \sim \text{Geom}(p)$,

$$P(X \geq j+k | X \geq j) = \frac{P(X \geq j+k)}{P(X \geq j)} = \frac{q^{j+k}}{q^j} = q^k = P(X \geq k).$$

Solution:



Let I_j be the indicator random variable for the j th box being empty, so $I_1 + \dots + I_n$ is the number of empty boxes (the above picture illustrates a possible outcome with 3 empty boxes, for $n = 9, k = 13$). Then $E(I_j) = P(I_j = 1) = (1 - 1/n)^k$. By linearity,

$$E\left(\sum_{j=1}^n I_j\right) = \sum_{j=1}^n E(I_j) = n(1 - 1/n)^k.$$

Solution: Creating an indicator r.v. for each pair of people, we have that the expected number of pairs of people with the same birthday is $\binom{50}{2} \frac{1}{365}$ by linearity. Now create an indicator r.v. for each day of the year, taking the value 1 if at least two of the people were born that day (and 0 otherwise). Then the expected number of days on which at least two people were born is $365 \left(1 - \left(\frac{364}{365}\right)^{50} - 50 \cdot \frac{1}{365} \cdot \left(\frac{364}{365}\right)^{49}\right)$.

Solution: The indicator I_{12} is independent of the indicator I_{34} since knowing the birthdays of persons 1 and 2 gives us no information about the birthdays of persons 3 and 4. Also, I_{12} is independent of I_{13} since even though both of these indicators involve person 1, knowing that persons 1 and 2 have the same birthday gives us no information about whether persons 1 and 3 have the same birthday (this relies on the assumption that the 365 days are equally likely). In general, the indicator r.v.s here are pairwise independent. But they are *not* independent since, for example, if person 1 has the same birthday as person 2 and person 1 has the same birthday as person 3, then persons 2 and 3 must have the same birthday.

Solution: Let X_j be the number of bags of gummi bears that the j th student gets, and let I_j be the indicator of $X_j \geq 1$. Then $X_j \sim \text{Bin}(20, \frac{1}{20})$, so $E(X_j) = 1$. So $E(X_1 + X_2 + X_3) = 3$ by linearity.

The average number of students who get at least one bag is

$$E(I_1 + \cdots + I_{20}) = 20E(I_1) = 20P(I_1 = 1) = 20 \left(1 - \left(\frac{19}{20} \right)^{20} \right).$$

Solution: Initially there are 200 free ends. The number of free ends decreases by 2 each time since either two separate pieces are tied together, or a new loop is formed. So exactly 100 steps are always needed. Let I_j be the indicator r.v. for whether a new loop is formed at the j th step. At the time when there are n unlooped pieces (so $2n$ ends), the probability of forming a new loop is $\frac{n}{\binom{2n}{2}} = \frac{1}{2n-1}$ since any 2 ends are equally likely to be chosen, and there are n ways to pick both ends of 1 of the n pieces. By linearity, the expected number of loops is

$$\sum_{n=1}^{100} \frac{1}{2n-1}.$$

Solution:

(a) In the story of the Hypergeometric, $\binom{X}{2}$ is the number of pairs of draws such that both balls are white. Creating an indicator r.v. for each pair, we have

$$E\binom{X}{2} = \binom{n}{2} \frac{w}{w+b} \frac{w-1}{w+b-1}.$$

(b) By (a),

$$EX^2 - EX = E(X(X-1)) = n(n-1)p \frac{w-1}{N-1},$$

so

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (EX)^2 \\ &= n(n-1)p \frac{w-1}{N-1} + np - n^2 p^2 \\ &= np \left(\frac{(n-1)(w-1)}{N-1} + 1 - np \right) \\ &= np \left(\frac{nw - w - n + N}{N-1} - \frac{nw}{N} \right) \\ &= np \left(\frac{Nnw - Nw - Nn + N^2 - Nnw + nw}{N(N-1)} \right) \\ &= np \left(\frac{(N-n)(N-w)}{N(N-1)} \right) \\ &= \frac{N-n}{N-1} npq. \end{aligned}$$

S16 Book Ch.4, Exercise 47

question · p. 95

Solution: Let I_j be an indicator random variable equal to 1 if the j^{th} location is empty, and 0 otherwise, for $1 \leq j \leq n$. Then $P(I_j = 1) = (1 - 1/n)^k$, since the phone numbers are stored in independent random locations. Then $I_1 + \dots + I_n$ is the number of empty locations. By linearity of expectation, we have

$$E\left(\sum_{j=1}^n I_j\right) = \sum_{j=1}^n E(I_j) = n(1 - 1/n)^k.$$

Similarly, the probability of a specific location having exactly 1 phone number stored is $\frac{k}{n}(1 - \frac{1}{n})^{k-1}$, so the expected number of such locations is $k(1 - 1/n)^{k-1}$. By linearity, the sum of the three expected values is n , so the expected number of locations with more than one phone number is $n - n(1 - 1/n)^k - k(1 - 1/n)^{k-1}$.

S17 Book Ch.4, Exercise 50

question · p. 95

Solution:

(a) There are $\binom{n}{2}$ pairs of numbers, each of which is equally likely to be in either order. So by symmetry, linearity, and indicator r.v.s, the expected number of inversions is $\frac{1}{2}\binom{n}{2}$.

(b) Let X be the number of comparisons and V be the number of inversions. On the one hand, $X \geq V$ since every inversion must be repaired. So $E(X) \geq E(V) = \frac{1}{2}\binom{n}{2}$. On the other hand, there are $n-1$ comparisons needed in the first sweep, $n-2$ in the second sweep (if needed), $\dots$, and 1 in the $(n-1)$ st sweep (if needed). So

$$X \leq (n-1) + (n-2) + \dots + 2 + 1 = \frac{n(n-1)}{2} = \binom{n}{2}.$$

Hence, $\frac{1}{2}\binom{n}{2} \leq E(X) \leq \binom{n}{2}$.

Solution:

(a) The distribution is $\text{Geom}(r)$, so the expected value is $\frac{1-r}{r}$.

(b) Use indicator random variables: let I_1 be 1 if green is obtained before red, and 0 otherwise, and define I_2 similarly for blue. Then

$$E(I_1) = P(\text{green before red}) = \frac{g}{g+r}$$

since “green before red” means that the first nonblue ball is green. Similarly, $E(I_2) = b/(b+r)$, so the expected number of colors obtained before getting red is

$$E(I_1 + I_2) = \frac{g}{g+r} + \frac{b}{b+r}.$$

(c) By definition of conditional probability,

$$P(\text{at least 2 red} | \text{at least 1 red}) = \frac{P(\text{at least 2 red})}{P(\text{at least 1 red})} = \frac{1 - (1-r)^n - nr(1-r)^{n-1}}{1 - (1-r)^n}.$$

Solution: For $n \geq 2$, $P(X > n)$ is the probability that none of $C_2, C_3, \dots, C_n$ are better candidates than C_1 , i.e., the probability that the first candidate is the highest ranked out of the first n . Since any ordering of the first n candidates is equally likely, each of the first n is equally likely to be the highest ranked of the first n , so $P(X > n) = 1/n$. For $n = 0$ or $n = 1$, $P(X > n) = 1$ (note that it does not make sense to say the probability is $1/n$ when $n = 0$). By Theorem 4.4.8,

$$E(X) = \sum_{n=0}^{\infty} P(X > n) = P(X > 0) + \sum_{n=1}^{\infty} P(X > n) = 1 + \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

since the series is the *harmonic series*, which diverges.

How can the average waiting time to find someone better than the first candidate be infinite? In the real world, there are always only finitely many candidates so the expected waiting time is finite, just as in the St. Petersburg paradox there must in reality be an upper bound on the number of rounds. The harmonic series diverges very slowly, so even with millions of job candidates the average waiting time would not be very large.

Solution: By LOTUS,

$$E(X!) = e^{-\lambda} \sum_{k=0}^{\infty} k! \frac{\lambda^k}{k!} = \frac{e^{-\lambda}}{1-\lambda},$$

for $0 < \lambda < 1$ since this is a geometric series (and $E(X!)$ is infinite if $\lambda \geq 1$).

Solution: Letting $q = 1 - p$, we have

$$E(e^{tX}) = p \sum_{k=0}^{\infty} e^{tk} q^k = p \sum_{k=0}^{\infty} (qe^t)^k = \frac{p}{1 - qe^t},$$

for $qe^t < 1$ (while for $qe^t \geq 1$, the series diverges).

Solution:

(a) We have $Y = X + 1$ with $X \sim \text{Pois}(\lambda)$, so $Y^2 = X^2 + 2X + 1$. So

$$E(Y^2) = E(X^2 + 2X + 1) = E(X^2) + 2E(X) + 1 = (\lambda + \lambda^2) + 2\lambda + 1 = \lambda^2 + 3\lambda + 1,$$

since $E(X^2) = \text{Var}(X) + (EX)^2 = \lambda + \lambda^2$.

(b) By LOTUS,

$$E\left(\frac{1}{Y}\right) = E\left(\frac{1}{X+1}\right) = \sum_{k=0}^{\infty} \frac{1}{k+1} e^{-\lambda} \frac{\lambda^k}{k!}.$$

Using the identity $k!(k+1) = (k+1)!$ and the Taylor series for e^λ , this becomes

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k+1)!} = \frac{e^{-\lambda}}{\lambda} \sum_{k=0}^{\infty} \frac{\lambda^{k+1}}{(k+1)!} = \frac{e^{-\lambda}}{\lambda} (e^\lambda - 1) = \frac{1}{\lambda} (1 - e^{-\lambda}).$$

Solution:

(a) The estimator is biased, with bias given by

$$\begin{aligned} E(e^{-3X}) - \theta &= \sum_{k=0}^{\infty} e^{-3k} \frac{\lambda^k}{k!} e^{-\lambda} - e^{-3\lambda} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(e^{-3}\lambda)^k}{k!} - e^{-3\lambda} \\ &= e^{-\lambda} e^{e^{-3}\lambda} - e^{-3\lambda} \\ &= e^{-3\lambda} (e^{(2+e^{-3})\lambda} - 1) \neq 0. \end{aligned}$$

(b) The estimator $g(X) = (-2)^X$ is unbiased since

$$\begin{aligned} E(-2)^X - \theta &= \sum_{k=0}^{\infty} (-2)^k \frac{\lambda^k}{k!} e^{-\lambda} - e^{-3\lambda} \\ &= e^{-\lambda} e^{-2\lambda} - e^{-3\lambda} = 0. \end{aligned}$$

(c) The estimator $g(X)$ is silly in the sense that it is sometimes negative, whereas $e^{-3\lambda}$ is positive. One simple way to get a better estimator is to modify $g(X)$ to make it nonnegative, by letting $h(X) = 0$ if $g(X) < 0$ and $h(X) = g(X)$ otherwise.

Better yet, note that $e^{-3\lambda}$ is between 0 and 1 since $\lambda > 0$, so letting $h(X) = 0$ if $g(X) < 0$ and $h(X) = 1$ if $g(X) > 0$ is clearly more sensible than using $g(X)$.

Solution:

(a) Let N be the number of students in the same seat for both classes. The problem has essentially the same structure as the matching problem. Let E_j be the event that the j th student sits in the same seat in both classes. Then

$$P(N = 0) = 1 - P\left(\bigcup_{j=1}^{100} E_j\right).$$

By symmetry, inclusion-exclusion gives

$$P\left(\bigcup_{j=1}^{100} E_j\right) = \sum_{j=1}^{100} (-1)^{j-1} \binom{100}{j} P\left(\bigcap_{k=1}^j E_k\right).$$

The j -fold intersection represents j particular students sitting pat throughout the two lectures, which occurs with probability $(100 - j)!/100!$. So

$$\begin{aligned} P\left(\bigcup_{j=1}^{100} E_j\right) &= \sum_{j=1}^{100} (-1)^{j-1} \binom{100}{j} \frac{(100 - j)!}{100!} = \sum_{j=1}^{100} (-1)^{j-1} / j! \\ P(N = 0) &= 1 - \sum_{j=1}^{100} \frac{(-1)^{j-1}}{j!} = \sum_{j=0}^{100} \frac{(-1)^j}{j!}. \end{aligned}$$

(b) Define I_i to be the indicator for student i having the same seat in both courses, so that $N = \sum_{i=1}^{100} I_i$. Then $P(I_i = 1) = 1/100$, and the I_i are weakly dependent because

$$P((I_i = 1) \cap (I_j = 1)) = \left(\frac{1}{100}\right) \left(\frac{1}{99}\right) \approx \left(\frac{1}{100}\right)^2 = P(I_i = 1)P(I_j = 1).$$

So N is close to $\text{Pois}(\lambda)$ in distribution, where $\lambda = E(N) = 100E(I_1) = 1$. Thus,

$$P(N = 0) \approx e^{-1} 1^0 / 0! = e^{-1} \approx 0.37.$$

This agrees with the result of (a), which we recognize as the Taylor series for e^x , evaluated at $x = -1$.

(c) Using a Poisson approximation, we have

$$P(N \geq 2) = 1 - P(N = 0) - P(N = 1) \approx 1 - e^{-1} - e^{-1} = 1 - 2e^{-1} \approx 0.26.$$

Solution:

(a) Let I_j be the indicator r.v. for the j th person picking his or her own name. Then $E(I_j) = P(I_j = 1) = \frac{1}{n}$. By linearity, the expected number is $n \cdot E(I_j) = 1$.

(b) Let I_{ij} be the indicator r.v. for the i th and j th persons having such a swap (for $i < j$). Then $E(I_{ij}) = P(i \text{ picks } j)P(j \text{ picks } i | i \text{ picks } j) = \frac{1}{n(n-1)}$.

Alternatively, we can get this by counting: there are $n!$ permutations for who picks whom, of which $(n-2)!$ have i pick j and j pick i , giving $\frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. So by linearity, the expected number is $\binom{n}{2} \cdot \frac{1}{n(n-1)} = \frac{1}{2}$.

(c) By the Poisson paradigm, X is approximately $\text{Pois}(1)$ for large n . As $n \rightarrow \infty$, $P(X = 0) \rightarrow 1/e$, which is the probability of a $\text{Pois}(1)$ r.v. being 0.

Solution:

(a) Let X be the number of people who win. Then

$$E(X) = \frac{10^7}{10^7} = 1.$$

A Poisson approximation is very good here since X is the number of “successes” for a very large number of independent trials where the probability of success on each trial is very low. So X is approximately $\text{Pois}(1)$, and for k a nonnegative integer,

$$P(X = k) \approx \frac{1}{e \cdot k!}.$$

(b) Let A be the event that you win the prize, and condition on W :

$$P(A) = \sum_{k=0}^{\infty} P(A|W = k)P(W = k) = \frac{1}{e} \sum_{k=0}^{\infty} \frac{1}{k+1} \frac{1}{k!} = \frac{1}{e} \sum_{k=0}^{\infty} \frac{1}{(k+1)!} = \frac{e-1}{e} = 1 - \frac{1}{e}.$$

Solution:

(a) Let I_j be the indicator for the j th movie being on both lists, for $1 \leq j \leq 11$. By symmetry and linearity, the desired expected value is

$$11 \left(\frac{5}{11} \right)^2 = \frac{25}{11}.$$

(b) Choose 2 *random* movies (one at a time, without replacement). Let X be the number of people who name both of these movies. Creating an indicator r.v. for each person,

$$E(X) = 111P(\text{Alice names both random movies}) = 111 \left(\frac{5}{11} \cdot \frac{4}{10} \right) = \left(\frac{111}{110} \right) 20 > 20,$$

since the first chosen movie has a $5/11$ chance of being on Alice’s list and given that it is, the second chosen movie has a $4/10$ chance of being on the list (or we can use the Hypergeometric PMF after “tagging” Alice’s favorite movies).

Thus, there must exist 2 movies such that at least 21 of the people name both of them as favorites.

Solution: Consider a random square, obtained by picking a uniformly random point on the circumference and inscribing a square with that point a corner; say that the corners are $U_1, \dots, U_4$, in clockwise order starting with the initial point chosen. Let I_j be the indicator r.v. of U_j touching red ink. By symmetry, $E(I_j) = 2/3$ so by linearity, the expected number of corners touching red ink is $8/3$. Thus, there must exist an inscribed square with at least $8/3$ of its corners touching red ink. Such a square must have at least 3 of its corners touching red ink.

Solution: Say that the “score” of a pair of students is how many problems at least one of them got right. The expected score of a random pair of students (with all pairs equally likely) is at least $8(1 - 0.35^2) = 7.02$, as seen by creating an indicator r.v. for each problem for the event that at least one student in the pair got it right. (We can also improve the 0.35^2 to $\frac{35}{100} \cdot \frac{34}{99}$ since the students are sampled without replacement.) So some pair of students must have gotten a score of at least 7.02, which means that they got a score of at least 8.

Solution: Take a uniformly random honeycomb tiling (to do this, start with any honeycomb tiling and then shift it horizontally and vertically by uniformly random amounts; by periodicity there is an upper bound on how large the shifts need to be). Choose the tiling so that a circle the same size as one of the coins can be inscribed in each hexagon. Then inscribe a circle in each hexagon, and let I_j be the indicator r.v. for the j th point being contained inside one of the circles. We have $E(I_j) > 0.9$ by the geometric fact mentioned above, so by linearity $E(I_1 + \cdots + I_{10}) > 9$. Thus, there is a positioning of the honeycomb tiling such that all 10 points are contained inside the circles. Putting coins on top of the circles containing the points, we can cover all ten points.

Solution: Generate m random strings of length n independently, using fair coin flips to determine each bit. Let S be the resulting random set of strings. If we can show that the probability that S is k -complete is *positive*, then we know that a k -complete set of size at most m must *exist*. Let A be the event that S is k -complete. Let $N = \binom{n}{k} 2^k$ and let $A_1, \dots, A_N$ be the events of the form “ S contains a string which is $b_1 \dots b_k$ at coordinates $i_1 < \cdots < i_k$,” in any fixed order. For example, if $k = 3$ then A_1 could be the event “ S has an element which is 110 at positions 1, 2, 3.” Then $P(A) > 0$ since

$$P(A^c) = P(\cup_{j=1}^N A_j^c) \leq \sum_{j=1}^N P(A_j^c) = N(1 - 2^{-k})^m < 1.$$

Solution:

(a) Imagine lining the people up and saying the first 3 are a team, the next 3 are a team, etc. This overcounts by a factor of $(3!)^{120} \cdot 120!$ since the order within teams and the order of teams don't matter. So the number of ways is

$$\frac{360!}{6^{120} \cdot 120!}.$$

(b) Let I_j be the indicator for the j th team having a married couple (taking the teams to be chosen one at a time, or with respect to a random ordering). By symmetry and linearity, the desired quantity is $120E(I_1)$. We have

$$E(I_1) = P(\text{first team has a married couple}) = \frac{180 \cdot 358}{\binom{360}{3}},$$

since the first team is equally likely to be any 3 of the people, and to have a married couple on the team we need to choose a couple and then any third person. So the expected value is

$$\frac{120 \cdot 180 \cdot 358}{\binom{360}{3}}.$$

(This simplifies to $\frac{120 \cdot 180 \cdot 358}{360 \cdot 359 \cdot 358/6} = \frac{360}{359}$. Another way to find the probability that the first team has a married couple is to note that any particular pair in the team has probability $\frac{1}{359}$ of being married to each other, so since there are 3 disjoint possibilities the probability is $\frac{3}{359}$.)

S33 Book Ch.4, Exercise 76

question · p. 99

Solution:

(a) Let I_j be the indicator r.v. for the event “all sixes” on the j th roll. Then $E(I_j) = 1/6^n$, so the expected value is $4 \cdot 6^{n-1}/6^n = 2/3$.

(b) By a Poisson approximation with $\lambda = 2/3$ (the expected value from (a)), the probability is approximately $1 - e^{-2/3}$.

(c) The answer stays the same, by the same reasoning as in (a), since linearity of expectation holds even for dependent r.v.s.

S34 Book Ch.4, Exercise 77

question · p. 99

Solution:

(a) Give each person \$10, and then distribute the remaining \$50 arbitrarily. By Bose-Einstein (thinking of people as boxes and dollars as balls!), the number of ways is

$$\binom{5 + 50 - 1}{50} = \binom{54}{50} = \binom{54}{4}.$$

(b) Let X_j be the amount that j gets. By symmetry, $E(X_j)$ is the same for all j . But $X_1 + \cdots + X_5 = 100$, so by linearity $100 = 5EX_1$. Thus, EX_1 is \$20.

(c) The events A_2 and A_3 are not independent since knowing that A_2 occurred makes it more likely that person 1 received a low percentage of the money, which in turn makes it more likely that A_3 occurred.

Solution:

(a) The distribution is $\text{Bin}(n, p)$ with $n = 11, p = \frac{1}{50}$ (thinking of getting a song from the favorite album as a “success”). So the PMF is

$$\binom{11}{k} \left(\frac{1}{50}\right)^k \left(\frac{49}{50}\right)^{11-k}, \text{ for } 0 \leq k \leq 11.$$

(b) This is a version of the birthday problem. We have

$$P(\text{at least 1 match}) = 1 - P(\text{no matches}) = 1 - \frac{50 \cdot 49 \cdot \dots \cdot 40}{50^{11}} = 1 - \frac{49!}{39! \cdot 50^{10}}.$$

(c) Defining an indicator r.v. I_{jk} for the event that the j th and k th songs match, we have $E(I_{jk}) = P(I_{jk} = 1) = 1/50$, so the expected number of matches is

$$\binom{11}{2} \frac{1}{50} = \frac{11 \cdot 10}{2 \cdot 50} = \frac{110}{100} = 1.1.$$

Solution:

(a) The distribution is Hypergeometric (think of capture-recapture, “tagging” the numbers you choose). So

$$\begin{aligned} P(\text{exactly 3 right} | \text{at least 1 right}) &= \frac{P(\text{exactly 3 right})}{1 - P(\text{none right})} \\ &= \frac{\binom{5}{3} \binom{30}{2} / \binom{35}{5}}{1 - \binom{5}{0} \binom{30}{5} / \binom{35}{5}}. \end{aligned}$$

(b) Let $n = \binom{35}{5}$. By the coupon collector problem (or directly by linearity, writing the expected number of days as a sum of T_j 's with $T_j - 1$ a Geometric), the expected value is

$$n \left(\frac{1}{n} + \frac{1}{n-1} + \dots + \frac{1}{2} + 1 \right).$$

(c) Let A_j be the event that j doesn't get picked, so

$$P(A_j) = (30/35)^{50} = (6/7)^{50}.$$

Let X be the number of A_j that occur. A Poisson approximation for X is reasonable since these events are rare and weakly dependent. This gives

$$P(X = 0) \approx e^{-35 \cdot (6/7)^{50}} \approx 0.98.$$

1. Find an example of two discrete random variables X and Y (on the same sample space) such that X and Y have the same distribution (i.e., same PMF and same CDF), but the event $X = Y$ *never* occurs.

For a simple example, let $X \sim \text{Bernoulli}(1/2)$ (i.e., X can be thought of as a fair coin flip), and let $Y = 1 - X$. Then Y is also $\text{Bernoulli}(1/2)$ by symmetry, but $X = Y$ is impossible. A more general example is to let $X \sim \text{Binomial}(n, 1/2)$ and $Y = n - X$, where n is any odd number (think of this as interchanging the definitions of “success” and “failure”).

S38 SP 4 §1 · Problem 2

question · p. 100

2. Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, etc. (so X takes values $1, 2, \dots, 7$, with equal probabilities). Let Y be the next day after X (again represented as an integer between 1 and 7). Do X and Y have the same distribution? What is $P(X < Y)$?

They have the same distribution since Y is also equally likely to represent any day of the week, but $P(X < Y) = P(X \neq 7) = \frac{6}{7}$.

S39 SP 4 §1 · Problem 3

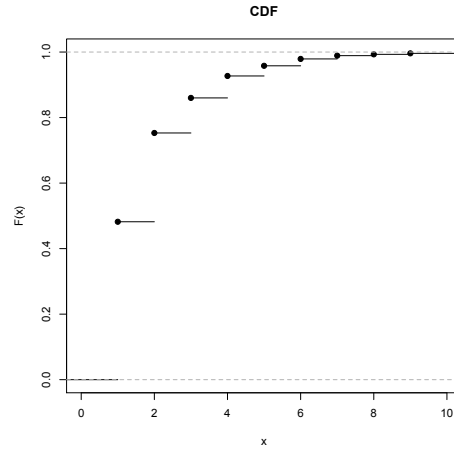
question · p. 100

3. A coin is tossed repeatedly until it lands Heads for the first time. Let X be the number of tosses that are required (including the toss that landed Heads), and let p be the probability of Heads. Find the CDF of X , and for $p = 1/2$ sketch its graph.

By the story of the Geometric, we have $X - 1 \sim \text{Geometric}(p)$. Using this or directly, the PMF is $P(X = k) = p(1 - p)^{k-1}$ for $k \in \{1, 2, 3, \dots\}$ (and 0 otherwise). The CDF can be obtained by adding up the PMF (from $k = 1$ to $k = \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x). We can also see directly that

$$P(X \leq x) = 1 - P(X > x) = 1 - (1 - p)^{\lfloor x \rfloor}$$

for $x \geq 1$, since $X > x$ says that the first $\lfloor x \rfloor$ flips land tails. The CDF is 0 for $x < 1$. For a fair coin, the CDF is $F(x) = 1 - \frac{1}{2^{\lfloor x \rfloor}}$ for $x \geq 1$, and $F(x) = 0$ for $x < 1$, as illustrated below.



S40 SP 4 §1 · Problem 4

question · p. 101

4. Are there discrete random variables X and Y such that $E(X) > 100E(Y)$ but Y is greater than X with probability at least 0.99?

Yes: consider what happens if we make X usually 0 but on rare occasions, X is extremely large (like the outcome of a lottery); Y , on the other hand, can be more moderate. For example, let X be 10^6 with probability $1/100$ and 0 with probability $99/100$, and let Y be the constant 1.

S41 SP 4 §1 · Problem 5

question · p. 101

5. Let X be a discrete r.v. with possible values $1, 2, 3, \dots$. Let $F(x) = P(X \leq x)$ be the CDF of X . Show that

$$E(X) = \sum_{n=0}^{\infty} (1 - F(n)).$$

Hint: organize the order of summation carefully, using the fact that, for example, $P(X > 3) = P(X = 4) + P(X = 5) + \dots$.

Note that

$$\sum_{n=0}^{\infty} (1 - F(n)) = \sum_{n=0}^{\infty} P(X > n) = \sum_{n=0}^{\infty} \sum_{k=n+1}^{\infty} P(X = k)$$

For each k , the term $P(X = k)$ appears exactly k times: there is one $P(X = k)$ term for each nonnegative integer $n < k$. More visually, write out some terms:

$$P(X = 1) + P(X = 2) + P(X = 3) + \dots +$$

$$\begin{aligned}
&P(X = 2) + P(X = 3) + P(X = 4) + \cdots + \\
&\quad P(X = 3) + P(X = 4) + \cdots + \\
&\quad \dots
\end{aligned}$$

Rearranging the terms of this series (which is allowed since the terms are non-negative),

$$\sum_{n=0}^{\infty} (1 - F(n)) = \sum_{k=1}^{\infty} kP(X = k) = E[X].$$

6. Job candidates $C_1, C_2, \dots$ are interviewed one by one, and the interviewer compares them and keeps an updated list of rankings (if n candidates have been interviewed so far, this is a list of the n candidates, from best to worst). Assume that there is no limit on the number of candidates available, that for any n the candidates $C_1, C_2, \dots, C_n$ are equally likely to arrive in any order, and that there are no ties in the rankings given by the interview.

Let X be the index of the first candidate to come along who ranks as better than the very first candidate C_1 (so C_X is better than C_1 , but the candidates after 1 but prior to X (if any) are worse than C_1). For example, if C_2 and C_3 are worse than C_1 but C_4 is better than C_1 , then $X = 4$. All $4!$ orderings of the first 4 candidates are equally likely, so it could have happened that the first candidate was the best out of the first 4 candidates, in which case $X > 4$.

What is $E(X)$ (which is a measure of how long, on average, the interviewer needs to wait to find someone better than the very first candidate)? Hint: find $P(X > n)$ by interpreting what $X > n$ says about how C_1 compares with other candidates, and then apply the result of the previous problem.

For $n \geq 2$, $P(X > n)$ is the probability that none of $C_2, C_3, \dots, C_n$ are better candidates than C_1 , i.e., the probability that the first candidate is the highest ranked out of the first n . Since any ordering of the first n candidates is equally likely, each of the first n is equally likely to be the highest ranked of the first n , so $P(X > n) = 1/n$. For $n = 0$ or $n = 1$, $P(X > n) = 1$ (note that it does not make sense to say the probability is $1/n$ when $n = 0$). Applying the result of the previous problem,

$$E(X) = \sum_{n=0}^{\infty} P(X > n) = P(X > 0) + \sum_{n=1}^{\infty} P(X > n) = 1 + \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

since the series is the *harmonic series*, which diverges.

How can the average waiting time to find someone better than the first candidate be infinite? In the real world, there are always only finitely many candidates so the expected waiting time is finite, just as in the St. Petersburg paradox there must in reality be an upper bound on the number of rounds. The harmonic series diverges very slowly, so even with millions of job candidates the average waiting time would not be very large.

S43 SP 4 §2 · Problem 1

[question · p. 101](#)

1. A group of 50 people are comparing their birthdays (as usual, assume their birthdays are independent, are not February 29, etc.). Find the expected number of pairs of people with the same birthday, and the expected number of days in the year on which at least two of these people were born.

Creating an indicator r.v. for each pair of people, we have that the expected number of pairs of people with the same birthday is $\binom{50}{2} \frac{1}{365}$ by linearity. Now create an indicator r.v. for each day of the year, taking the value 1 if at least two of the people were born that day (and 0 otherwise). Then the expected number of days on which at least two people were born is $365 \left(1 - \left(\frac{364}{365}\right)^{50} - 50 \cdot \frac{1}{365} \cdot \left(\frac{364}{365}\right)^{49}\right)$.

S44 SP 4 §2 · Problem 2

[question · p. 102](#)

2. A total of 20 bags of Haribo gummi bears are randomly distributed to the 20 students in a certain Stat 110 section. Each bag is obtained by a random student, and the outcomes of who gets which bag are independent. Find the average number of bags of gummi bears that the first three students get in total, and find the average number of students who get at least one bag.

Let X_j be the number of bags of gummi bears that the j th student gets, and let I_j be the indicator of $X_j \geq 1$. Then $X_j \sim \text{Bin}(20, \frac{1}{20})$, so $E(X_j) = 1$. So $E(X_1 + X_2 + X_3) = 3$ by linearity.

The average number of students who get at least one bag is

$$E(I_1 + \cdots + I_{20}) = 20E(I_1) = 20P(I_1 = 1) = 20 \left(1 - \left(\frac{19}{20}\right)^{20}\right).$$

S45 SP 4 §2 · Problem 3

[question · p. 102](#)

3. There are 100 shoelaces in a box. At each stage, you pick two random ends and tie them together. Either this results in a longer shoelace (if the two ends came from different pieces), or it results in a loop (if the two ends came from the same piece). What are the expected number of steps until everything is in loops, and the expected number of loops after everything is in loops? (This is a famous interview problem; leave the latter answer as a sum.)

Hint: for each step, create an indicator r.v. for whether a loop was created then, and note that the number of free ends goes down by 2 after each step.

Initially there are 200 free ends. The number of free ends decreases by 2 each time since either two separate pieces are tied together, or a new loop is formed. So exactly 100 steps are always needed. Let I_j be the indicator r.v. for whether a new loop is formed at the j th step. At the time when there are n unlooped pieces (so $2n$ ends), the probability of forming a new loop is $\frac{n}{\binom{2n}{2}} = \frac{1}{2n-1}$ since any 2 ends are equally likely to be chosen, and there are n ways to pick both ends of 1 of the n pieces. By linearity, the expected number of loops is

$$\sum_{n=1}^{100} \frac{1}{2n-1}.$$

4. A *hash table* is a commonly used data structure in computer science, allowing for fast information retrieval. For example, suppose we want to store some people's phone numbers. Assume that no two of the people have the same name. For each name x , a *hash function* h is used, where $h(x)$ is the location to store x 's phone number. After such a table has been computed, to look up x 's phone number one just recomputes $h(x)$ and then looks up what is stored in that location.

The hash function h is deterministic, since we don't want to get different results every time we compute $h(x)$. But h is often chosen to be *pseudorandom*. For this problem, assume that true randomness is used. So let there be k people, with each person's phone number stored in a random location (independently), represented by an integer between 1 and n . It then might happen that one location y has more than one phone number stored there, if two different people x and y end up with the same random location for their information to be stored.

Find the expected number of locations with no phone numbers stored, the

expected number with exactly one phone number, and the expected number with more than one phone number (should these quantities add up to n ?).

Let I_j be an indicator random variable equal to 1 if the j^{th} location is empty, and 0 otherwise, for $1 \leq j \leq n$. Then $P(I_j = 1) = (1 - 1/n)^k$, since the phone numbers are stored in independent random locations. Then $I_1 + \cdots + I_n$ is the number of empty locations. By linearity of expectation, we have

$$E\left(\sum_{j=1}^n I_j\right) = \sum_{j=1}^n E(I_j) = n(1 - 1/n)^k.$$

Similarly, the probability of a specific location having exactly 1 phone number stored is $\frac{k}{n}(1 - \frac{1}{n})^{k-1}$, so the expected number of such locations is $k(1 - 1/n)^{k-1}$. By linearity, the sum of the three expected values is n , so the expected number of locations with more than one phone number is $n - n(1 - 1/n)^k - k(1 - 1/n)^{k-1}$.

S47 SP 5 §1 · Problem 1

[question · p. 103](#)

1. Raindrops are falling at an average rate of 20 drops per square inch per minute. What would be a reasonable distribution to use for the number of raindrops hitting a particular region measuring 5 inches² in t minutes? Why? Using your chosen distribution, compute the probability that the region has no rain drops in a given 3 second time interval.

A reasonable choice of distribution is Poisson(λt), where $\lambda = 20 \cdot 5 = 100$ (the average number of raindrops per minute hitting the region). Assuming this distribution,

$$P(\text{no raindrops in } 1/20 \text{ of a minute}) = e^{-100/20}(100/20)^0/0! = e^{-5}.$$

S48 SP 5 §1 · Problem 2

[question · p. 103](#)

2. Harvard Law School courses often have assigned seating to facilitate the “Socratic method.” Suppose that there are 100 first year Harvard Law students, and each takes two courses: Torts and Contracts. Both are held in the same lecture hall (which has 100 seats), and the seating is uniformly random and independent for the two courses.

(a) Find the probability that no one has the same seat for both courses (exactly; you should leave your answer as a sum).

Let N be the number of students in the same seat for both classes. The problem has the same structure as the *de Montmort matching problem* from lecture. Let E_j be the event that the j^{th} student sits in the same seat in both classes. Then

$$P(N = 0) = 1 - P\left(\bigcup_{j=1}^{100} E_j\right)$$

By symmetry, inclusion-exclusion gives

$$P\left(\bigcup_{j=1}^{100} E_j\right) = \sum_{j=1}^{100} (-1)^{j-1} \binom{100}{j} P\left(\bigcap_{k=1}^j E_k\right)$$

The j -term intersection event represents j particular students sitting pat throughout the two lectures, which occurs with probability $(100 - j)!/100!$. So

$$P\left(\bigcup_{j=1}^{100} E_j\right) = \sum_{j=1}^{100} (-1)^{j-1} \binom{100}{j} \frac{(100 - j)!}{100!} = \sum_{j=1}^{100} (-1)^{j-1} / j!$$

$$P(N = 0) = 1 - \sum_{j=1}^{100} \frac{(-1)^{j-1}}{j!} = \sum_{j=0}^{100} \frac{(-1)^j}{j!}.$$

(b) Find a simple but accurate approximation to the probability that no one has the same seat for both courses.

Define I_i to be the indicator for student i having the same seat in both courses, so that $N = \sum_{i=1}^{100} I_i$. Then $P(I_i = 1) = 1/100$, and the I_i are weakly dependent because

$$P((I_i = 1) \cap (I_j = 1)) = \left(\frac{1}{100}\right) \left(\frac{1}{99}\right) \approx \left(\frac{1}{100}\right)^2 = P(I_i = 1)P(I_j = 1)$$

So N is close to $\text{Pois}(\lambda)$ in distribution, where $\lambda = E(N) = 100EI_1 = 1$. Thus,

$$P(N = 0) \approx e^{-1} 1^0 / 0! = e^{-1}.$$

This agrees with the result of (a), which we recognize as the Taylor series for e^x , evaluated at $x = -1$.

(c) Find a simple but accurate approximation to the probability that at least two students have the same seat for both courses.

Using a Poisson approximation, we have

$$P(N \geq 2) = 1 - P(N = 0) - P(N = 1) \approx 1 - e^{-1} - e^{-1} = 1 - 2e^{-1}.$$

3. Let X be a $\text{Pois}(\lambda)$ random variable, where λ is fixed but unknown. Let $\theta = e^{-3\lambda}$, and suppose that we are interested in estimating θ based on the data. Since X is what we observe, our estimator is a function of X , call it $g(X)$. The *bias* of the estimator $g(X)$ is defined to be $E(g(X)) - \theta$, i.e., how far off the estimate is on average; the estimator is *unbiased* if its bias is 0.

(a) For estimating λ , the r.v. X itself is an unbiased estimator. Compute the bias of the estimator $T = e^{-3X}$. Is it unbiased for estimating θ ?

The estimator is biased, with bias given by

$$\begin{aligned}
 E(e^{-3X}) - \theta &= \sum_{k=0}^{\infty} e^{-3k} \frac{\lambda^k}{k!} e^{-\lambda} - e^{-3\lambda} \\
 &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(e^{-3}\lambda)^k}{k!} - e^{-3\lambda} \\
 &= e^{-\lambda} e^{e^{-3}\lambda} - e^{-3\lambda} \\
 &= e^{-3\lambda} (e^{(2+e^{-3})\lambda} - 1) \neq 0.
 \end{aligned}$$

(b) Show that $g(X) = (-2)^X$ is an unbiased estimator for θ . (In fact, it is the best unbiased estimator, in the sense of minimizing the average squared error.)

The estimator $g(X) = (-2)^X$ is unbiased since

$$\begin{aligned}
 E(-2)^X - \theta &= \sum_{k=0}^{\infty} (-2)^k \frac{\lambda^k}{k!} e^{-\lambda} - e^{-2\lambda} \\
 &= e^{-\lambda} e^{-2\lambda} - e^{-3\lambda} = 0.
 \end{aligned}$$

(c) Explain intuitively why $g(X)$ is a silly choice for estimating θ , despite (b), and show how to improve it by finding an estimator $h(X)$ for θ that is always at least as good as $g(X)$ and sometimes strictly better than $g(X)$. That is,

$$|h(X) - \theta| \leq |g(X) - \theta|,$$

with the inequality sometimes strict.

The estimator $g(X)$ is silly in the sense that it is sometimes negative, whereas $e^{-3\lambda}$ is positive. One simple way to get a better estimator is to modify $g(X)$ to make it nonnegative, by letting $h(X) = 0$ if $g(X) < 0$ and $h(X) = g(X)$ otherwise.

Better yet, note that $e^{-3\lambda}$ is between 0 and 1 since $\lambda > 0$, so letting $h(X) = 0$ if $g(X) < 0$ and $h(X) = 1$ if $g(X) > 0$ is clearly more sensible than using $g(X)$. It turns out that the answer to this part *must* be biased: it can be shown that $g(X)$ is the *only* unbiased estimator for θ , even though it would be silly to use $g(X)$ in practice.

1. Let $Z \sim \mathcal{N}(0, 1)$ and let S be a “random sign” independent of Z , i.e., S is 1 with probability $1/2$ and -1 with probability $1/2$. Show that $SZ \sim \mathcal{N}(0, 1)$.

Condition on S to find the CDF of SZ :

$$\begin{aligned}
 P(SZ \leq x) &= P(SZ \leq x | S = 1) \frac{1}{2} + P(SZ \leq x | S = -1) \frac{1}{2} \\
 &= P(Z \leq x) \frac{1}{2} + P(Z \geq -x) \frac{1}{2} \\
 &= P(Z \leq x) \frac{1}{2} + P(Z \leq x) \frac{1}{2} \\
 &= \Phi(x),
 \end{aligned}$$

where the penultimate equality is by symmetry of the Normal.

S51 SP 5 §2 · Problem 2

question · p. 104

2. Explain why $P(X < Y) = P(Y < X)$ if X and Y are i.i.d. Does it follow that $P(X < Y) = 1/2$? Is it still always true that $P(X < Y) = P(Y < X)$ if X and Y have the same distribution but are not independent?

If X and Y are i.i.d., then $P(X < Y) = P(Y < X)$ by symmetry: we can interchange X and Y since both are the probability of one draw from the distribution being less than another, independent draw. In the discrete case, $P(X < Y) < 1/2$ since $P(X = Y) > 0$. In the continuous case, $P(X < Y) = 1/2$ since $P(X = Y) = 0$. If X and Y are not independent, then it is not necessarily true that $P(X < Y) = P(Y < X)$ since they may be structured in a way that tends to make X less than Y .

S52 SP 5 §2 · Problem 3

question · p. 104

3. Explain why if $X \sim \text{Bin}(n, p)$, then $n - X \sim \text{Bin}(n, 1 - p)$.

This follows immediately from the story of the Binomial, by redefining “success” as “failure” and vice versa.

S53 SP 5 §2 · Problem 4

question · p. 104

4. There are 100 passengers lined up to board an airplane with 100 seats (with each seat assigned to one of the passengers). The first passenger in line crazily decides to sit in a randomly chosen seat (with all seats equally likely). Each subsequent passenger takes his or her assigned seat if available, and otherwise sits in a random available seat. What is the probability that the last passenger in line gets to sit in his or her assigned seat? (This is another common interview problem, and a beautiful example of the power of symmetry.)

Hint: call the seat assigned to the j th passenger in line “Seat j ” (regardless of whether the airline calls it seat 23A or whatever). What are the possibilities for which seats are available to the last passenger in line, and what is the probability of each of these possibilities?

The seat for the last passenger is either Seat 1 or Seat 100; for example, Seat 42 can't be available to the last passenger since the 42nd passenger in line would have sat there if possible. Seat 1 and Seat 100 are equally likely to be available to the last passenger, since the previous 99 passengers view these two seats symmetrically. So the probability that the last passenger gets Seat 100 is $1/2$.

S54 SP 8 §3 · Problem 1

question · p. 104

1. Let S be a set of binary strings $a_1 \dots a_n$ of length n (where juxtaposition means concatenation). We call S k -complete if for any indices $1 \leq i_1 < \dots < i_k \leq n$ and any binary string $b_1 \dots b_k$ of length k , there is a string $s_1 \dots s_n$ in S such that $s_{i_1} s_{i_2} \dots s_{i_k} = b_1 b_2 \dots b_k$. For example, for $n = 3$, the set $S = \{001, 010, 011, 100, 101, 110\}$ is 2-complete since all 4 patterns of 0's and 1's of length 2 can be found in any 2 positions. Show that if $\binom{n}{k} 2^k (1 - 2^{-k})^m < 1$, then there exists a k -complete set of size at most m .

Generate m random strings of length n independently, using fair coin flips to determine each bit. Let S be the resulting random set of strings. If we can show that the probability that S is k -complete is *positive*, then we know that a k -complete set of size at most m must *exist*. Let A be the event that S is k -complete. Let $N = \binom{n}{k} 2^k$ and let $A_1, \dots, A_N$ be the events of the form “ S contains a string which is $b_1 \dots b_k$ at coordinates $i_1 < \dots < i_k$,” in any fixed order. For example, if $k = 3$ then A_1 could be the event “ S has an element which is 110 at positions 1, 2, 3.” Then $P(A) > 0$ since

$$P(A^c) = P(\cup_{j=1}^N A_j^c) \leq \sum_{j=1}^N P(A_j^c) = N(1 - 2^{-k})^m < 1.$$

S55 SP 8 §3 · Problem 2

question · p. 105

2. A hundred students have taken an exam consisting of 8 problems, and for each problem at least 65 of the students got the right answer. Show that there exist two students who collectively got everything right, in the sense that for each problem, at least one of the two got it right.

Say that the “score” of a pair of students is how many problems at least one of them got right. The expected score of a random pair of students (with all pairs equally likely) is at least $8(1 - 0.35^2) = 7.02$, as seen by creating an indicator r.v. for each problem for the event that at least one student in the pair got it right. (We can also improve the 0.35^2 to $\frac{35}{100} \cdot \frac{34}{99}$ since the students are sampled without replacement.) So some pair of students must have gotten a score of at least 7.02, which means that they got a score of 8! ($\leftarrow$ not a factorial.)

S56 SP 8 §3 · Problem 3

question · p. 105

3. The circumference of a circle is colored with red and blue ink such that $2/3$ of the circumference is red and $1/3$ is blue. Prove that no matter how complicated the coloring scheme is, there is a way to inscribe a square in the circle such that at least three of the four corners of the square touch red ink.

Consider a random square, obtained by picking a uniformly random point on the circumference and inscribing a square with that point a corner; say that the corners are $U_1, \dots, U_4$, in clockwise order starting with the initial point chosen. Let I_j be the indicator r.v. of U_j touching red ink. By symmetry, $E(I_j) = 2/3$ so by linearity, the expected number of corners touching red ink is $8/3$. Thus, there must exist an inscribed square with at least $8/3$ of its corners touching red ink. Such a square must have at least 3 of its corners touching red ink.

S57 SP 8 §3 · Problem 4

question · p. 105

4. Ten points in the plane are designated. You have ten circular coins (of the same radius). Show that you can position the coins in the plane (without stacking them) so that all ten points are covered.

Hint: consider a honeycomb tiling as in <http://mathworld.wolfram.com/Honeycomb>. You can use the fact from geometry that if a circle is inscribed in a hexagon then the ratio of the area of the circle to the area of the hexagon is $\frac{\pi}{2\sqrt{3}} > 0.9$.

Take a uniformly random honeycomb tiling (to do this, start with any honeycomb tiling and then shift it horizontally and vertically by uniformly random amounts; by periodicity there is a bound on how large the shifts need to be). Choose the tiling so that a circle the same size as one of the coins can be inscribed in each hexagon. Then inscribe a circle in each hexagon, and let I_j be the indicator r.v. for the j th point being contained inside one the circles.

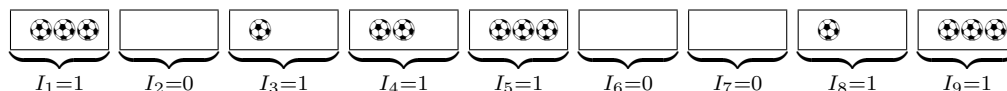
We have $E(I_j) > 0.9$ by the geometric fact mentioned above, so by linearity $E(I_1 + \dots + I_{10}) > 9$. Thus, there is a positioning of the honeycomb tiling such that all 10 points are contained inside the circles. Putting coins on top of the circles containing the points, we can cover all ten points.

3. A couple decides to keep having children until they have at least one boy and at least one girl, and then stop. Assume they never have twins, that the “trials” are independent with probability $1/2$ of a boy, and that they are fertile enough to keep producing children indefinitely. What is the expected number of children?

Let X be the number of children needed, starting with the 2nd child, to obtain one whose gender is not the same as that of the firstborn. Then $X - 1$ is $\text{Geom}(1/2)$, so $E(X) = 2$. This does not include the firstborn, so the expected total number of children is $E(X + 1) = E(X) + 1 = 3$.

Miracle check: an answer of 2 or lower would be a miracle since the couple always needs to have at least 2 children, and sometimes they need more. An answer of 4 or higher would be a miracle since 4 is the expected number of children needed such that there is a boy and a girl with the boy older than the girl.

4. Randomly, k distinguishable balls are placed into n distinguishable boxes, with all possibilities equally likely. Find the expected number of empty boxes.



Let I_j be the indicator random variable for the j^{th} box being empty, so $I_1 + \cdots + I_n$ is the number of empty boxes (the above picture illustrates a possible outcome with $n = 9, k = 13$). Then $E(I_j) = P(I_j = 1) = (1 - 1/n)^k$. By linearity,

$$E\left(\sum_{j=1}^n I_j\right) = \sum_{j=1}^n E(I_j) = n(1 - 1/n)^k.$$

Miracle check: for any $k \geq 1$, there can be at most $n - 1$ empty boxes, so the expected number of empty boxes must be at most $n - 1$. Here, we do have $n(1 - 1/n)^k \leq n(1 - 1/n) = n - 1$. And for $k = 0$ the answer should reduce to n , while for $n = 1, k \geq 1$ it should reduce to 0. Also, it makes sense that the expected number of empty boxes converges to 0 (without ever reaching 0, if $n \geq 2$) as $k \rightarrow \infty$.

6. Consider the following algorithm for sorting a list of n distinct numbers into increasing order. Initially they are in a random order, with all orders equally likely. The algorithm compares the numbers in positions 1 and 2, and swaps them if needed, then it compares the new numbers in positions 2 and 3, and swaps them if needed, etc., until it has gone through the whole list. Call this one “sweep” through the list. After the first sweep, the largest number is at the end, so the second sweep (if needed) only needs to work with the first $n - 1$ positions. Similarly, the third sweep (if needed) only needs to work with the first $n - 2$ positions, etc. Sweeps are performed until $n - 1$ sweeps have been completed or there is a swapless sweep.

For example, if the initial list is 53241 (omitting commas), then the following 4 sweeps are performed to sort the list, with a total of 10 comparisons:

$$53241 \rightarrow 35241 \rightarrow 32541 \rightarrow 32451 \rightarrow 32415.$$

$$32415 \rightarrow 23415 \rightarrow 23415 \rightarrow 23145.$$

$$23145 \rightarrow 23145 \rightarrow 21345.$$

$$21345 \rightarrow 12345.$$

(a) An *inversion* is a pair of numbers that are out of order (e.g., 12345 has no inversions, while 53241 has 8 inversions). Find the expected number of inversions in the original list.

There are $\binom{n}{2}$ pairs of numbers, each of which is equally likely to be in either order. So by symmetry, linearity, and indicator r.v.s, we immediately have that the expected number of inversions is $\frac{1}{2}\binom{n}{2}$.

(b) Show that the expected number of comparisons is between $\frac{1}{2}\binom{n}{2}$ and $\binom{n}{2}$.

Hint for (b): for one bound, think about how many comparisons are made if $n - 1$ sweeps are done; for the other bound, use Part (a).

Let X be the number of comparisons and V be the number of inversions. On the one hand, $X \geq V$ since every inversion must be repaired. So $E(X) \geq E(V) = \frac{1}{2}\binom{n}{2}$. On the other hand, there are $n - 1$ comparisons needed in the first sweep, $n - 2$ in the second sweep (if needed), $\dots$, and 1 in the $(n - 1)$ st sweep (if needed). So

$$X \leq (n - 1) + (n - 2) + \dots + 2 + 1 = \frac{n(n - 1)}{2} = \binom{n}{2}.$$

Hence, $\frac{1}{2}\binom{n}{2} \leq E(X) \leq \binom{n}{2}$. This algorithm is known as *bubble-sort*. See <http://www.youtube.com/watch?v=lyZQPjUT5B4> for a Hungarian folk dance of the bubble-sort algorithm!

7. Athletes compete one at a time at the high jump. Let X_j be how high the j th jumper jumped, with $X_1, X_2, \dots$ i.i.d. with a continuous distribution. We say that the j th jumper set a *record* if X_j is greater than all of $X_{j-1}, \dots, X_1$.

(a) Is the event “the 110th jumper sets a record” independent of the event “the 111th jumper sets a record”? Justify your answer by finding the relevant probabilities in the definition of independence *and* with an intuitive explanation.

Let I_j be the indicator r.v. for the j th jumper setting a record. By symmetry, $P(I_j = 1) = 1/j$ (as all of the first j jumps are equally likely to be the highest of those jumps). Also,

$$P(I_{110} = 1, I_{111} = 1) = \frac{109!}{111!} = \frac{1}{110 \cdot 111},$$

since having the 110th and 111th jumps both being records is the same thing as having the highest of the first 111 jumps being in position 111, the second highest being in position 110, and the remaining 109 being in any order. So

$$P(I_{110} = 1, I_{111} = 1) = P(I_{110} = 1)P(I_{111} = 1),$$

which shows that the 110th jumper setting a record is independent of the 111th jumper setting a record. Intuitively, this makes sense since learning that the 111th jumper sets a record gives us no information about the “internal” matter of how the first 110 jumps are arranged amongst themselves.

(b) Find the mean number of records among the first n jumpers (as a sum). What happens to the mean as $n \rightarrow \infty$?

By linearity, the expected number of records among the first n jumpers is $\sum_{j=1}^n \frac{1}{j}$, which goes to ∞ as $n \rightarrow \infty$ (as this is the harmonic series).

1. A group of $n \geq 4$ people are comparing their birthdays (as usual, assume their birthdays are independent, are not February 29, etc.).

(a) Let I_{ij} be the indicator r.v. of i and j having the same birthday (for $i < j$). Is I_{12} independent of I_{34} ? Is I_{12} independent of I_{13} ? Are the I_{ij} 's independent?

The indicator I_{12} is independent of the indicator I_{34} since knowing the birthdays of persons 1 and 2 gives us no information about the birthdays of persons 3 and 4. Also, I_{12} is independent of I_{13} since even though both of these indicators involve person 1, knowing that persons 1 and 2 have the same birthday gives us no information about whether persons 1 and 3 have the same birthday (this relies on the assumption that the 365 days are equally likely). In general, the indicator r.v.s here are pairwise independent. But they are *not* independent since, for example, if person 1 has the same birthday as person 2 and person 1 has the same birthday as person 3, then persons 2 and 3 must have the same birthday.

(b) Explain why the Poisson Paradigm is applicable here even for moderate n , and use it to get a good approximation to the probability of at least 1 match when $n = 23$.

Let X be the number of birthday matches in a group of n people. There are at most $\binom{n}{2}$ matches, so the possible values of X are $0, 1, \dots, \binom{n}{2}$. The Poisson paradigm says that if we have a large number n of events which are independent (or weakly dependent), each of which has a small probability of occurring, then how many of these events occur will be approximately Poisson.

In the birthday problem, even for moderate n , the number of experiments is much larger: $\binom{n}{2} \approx n^2/2$, the experiments are weakly dependent (in fact they are pairwise independent), and the probabilities for each event to occur are small (each of the events has probability $\frac{1}{365}$ of occurring). Using a Poisson approximation, X is approximately $\text{Pois}(\frac{n(n-1)}{730})$ in distribution, which is $\text{Pois}(0.693)$ for $n = 23$. Then

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &\approx 1 - e^{-0.693} \\ &\approx 0.500. \end{aligned}$$

(c) About how many people are needed so that there is a 50% chance (or better) that two either have the same birthday or are only 1 day apart? (Note that this is much harder than the birthday problem to do exactly, but the Poisson Paradigm makes it possible to get fairly accurate approximations quickly.)

Now define the r.v. I_{ij} to be the indicator of “persons i and j have the same birthday or are only one day apart”. So $P(I_{ij} = 1) = \frac{3}{365}$. Again we can use the Poisson paradigm, letting X be the number of “near-matches”. By Poisson approximation, X is approximately a $\text{Pois}(\frac{3n(n-1)}{730})$. Now set the Poisson approximation for $P(X \geq 1)$ equal to $1/2$, and solve for n :

$$\begin{aligned} P(X \geq 1) = 1 - P(X = 0) &\approx 1 - e^{\frac{-3n^2+3n}{730}} = 1/2 \\ \frac{-3n^2+3n}{730} &= -\log 2 \\ 3n^2 - 3n - 730 \log 2 &= 0. \end{aligned}$$

Solving this quadratic equation, we obtain $n \approx 13.5$. So with 14 people, there should be a slightly better than 50-50 chance of two having the same birthday or birthdays one day apart. (The exact answer for the number of people needed turns out to be $n = 14$, through much harder and more tedious calculations; so the Poisson approximation works very well here!)

3. Consider an experiment where we observe the value of a random variable X , and estimate the value of an unknown constant θ using some random variable $T = g(X)$ that is a function of X . The r.v. T is called an *estimator*. Think of X as the data observed in the experiment, and θ as an unknown parameter related to the distribution of X .

For example, consider the experiment of flipping a coin n times, where the coin has an unknown probability θ of Heads. After the experiment is performed, we have observed the value of $X \sim \text{Bin}(n, \theta)$. The most natural estimator for θ is then X/n .

(a) The *bias* of an estimator T for θ is defined as $b(T) = E(T) - \theta$. The *mean squared error* is the average squared error when using $T(X)$ to estimate θ :

$$\text{MSE}(T) = E(T - \theta)^2.$$

Show that

$$\text{MSE}(T) = \text{Var}(T) + (b(T))^2.$$

This implies that for fixed MSE, lower bias can only be attained at the cost of higher variance and vice versa; this is a form of the *bias-variance tradeoff*, a phenomenon which arises throughout statistics.

Using the fact that adding a constant does not affect variance, we have

$$\begin{aligned}\text{Var}(T) &= \text{Var}(T - \theta) \\ &= E(T - \theta)^2 - (E(T - \theta))^2 \\ &= \text{MSE}(T) - (b(T))^2,\end{aligned}$$

which proves the desired identity.

(b) Show without using calculus that the constant c that minimizes $E(X - c)^2$ is the expected value of X . (This means that in choosing a single number to summarize X , the mean is the best choice if the goal is to minimize the average squared error.)

Applying (a) (with c in place of θ and $T(X) = X$),

$$E(X - c)^2 = \text{Var}(X) + (E(X) - c)^2 \geq \text{Var}(X)$$

for all c . Equality holds if and only if $E(X) - c = 0$, which is equivalent to $c = EX$. Thus, the unique value of c that minimizes $E(X - c)^2$ is EX . Note that for $c = EX$, the value of $E(X - c)^2$ is $\text{Var}(X)$.

(c) For the case that X is continuous with PDF $f(x)$ which is positive everywhere, show that the value of c that minimizes $E|X - c|$ is the median of X (which is the value m with $P(X \leq m) = 1/2$).

Hint: this can be done either with or without calculus. For the calculus method, use LOTUS to write $E|X - c|$ as an integral, and then split the integral into 2 pieces to get rid of the absolute values. Then use the fundamental theorem of calculus (after writing, for example, $\int_{-\infty}^c (c - x)f(x)dx = c \int_{-\infty}^c f(x)dx - \int_{-\infty}^c xf(x)dx$).

Proof with calculus: We want to minimize

$$E|X - c| = \int_{-\infty}^{\infty} |x - c|f(x)dx = \int_{-\infty}^c (c - x)f(x)dx + \int_c^{\infty} (x - c)f(x)dx,$$

where we split the integral into 2 pieces to handle the absolute values. This becomes

$$E|X - c| = c \int_{-\infty}^c f(x)dx - \int_{-\infty}^c xf(x)dx + \int_c^{\infty} xf(x)dx - c \int_c^{\infty} f(x)dx.$$

Now differentiate both sides with respect to c , using the fundamental theorem of calculus:

$$\frac{d}{dc}(E|X - c|) = \int_{-\infty}^c f(x)dx + cf(c) - cf(c) - cf(c) - \int_c^{\infty} f(x)dx + cf(c),$$

which simplifies to $P(X \leq c) - (1 - P(X \leq c)) = 2P(X \leq c) - 1$. This is 0 when $P(X \leq c) = 1/2$. The second derivative is $f(c) + f(c) = 2f(c) > 0$, so we have found a minimum. Thus, $E|X - c|$ is minimized when c is the median.

Proof without calculus: We will show a more general result, not assuming that X is a continuous r.v. A number m is called a *median* of the distribution of X if $P(X \leq m) \geq 1/2, P(X \geq m) \geq 1/2$. (So a median may not be unique; it will be unique if the CDF is continuous and strictly increasing.) Let m be a median, and let $a \neq m$. We need to show $E|X - m| \leq E|X - a|$, which is equivalent to $E(|X - a| - |X - m|) \geq 0$. Assume $m < a$ (the case $m > a$ can be handled by the same method).

Note that $|X - a| - |X - m| = a - X - (m - X) = a - m$ if $X \leq m$, and $|X - a| - |X - m| \geq X - a - (X - m) = m - a$ if $X > m$. Splitting the definition of expected value into 2 parts based on whether $X > m$ occurs, we have

$$E(|X - a| - |X - m|) \geq (a - m)P(X \leq m) + (m - a)P(X > m),$$

which simplifies to $(a - m)(P(X \leq m) - P(X > m))$. By definition of median,

$$P(X \leq m) - P(X > m) = P(X \leq m) - (1 - P(X \leq m)) = 2P(X \leq m) - 1 \geq 0,$$

which shows that $E(|X - m|) \leq E(|X - a|)$.

7. Shakespeare wrote a total of 884647 words in his known works. Of course, many words are used more than once, and the number of distinct words in Shakespeare's known writings is 31534 (according to one computation). This puts a lower bound on the size of Shakespeare's vocabulary, but it is likely that Shakespeare knew words which he did not use in these known writings.

More specifically, suppose that a new poem of Shakespeare were uncovered, and consider the following (seemingly impossible) problem: give a good prediction of the number of words in the new poem that do not appear anywhere in Shakespeare's previously known works.

The statisticians Ronald Thisted and Bradley Efron studied this problem in a paper called "Did Shakespeare write a newly-discovered poem?", which performed statistical tests to try to determine whether Shakespeare was the author of a poem discovered by a Shakespearean scholar in 1985. A simplified version of their method is developed in the problem below. The method was originally invented by Alan Turing (the founder of computer science) and I.J. Good as part of the effort to break the German Enigma code during World War II.

Let N be the number of distinct words that Shakespeare knew, and assume these words are numbered from 1 to N . Suppose for simplicity that Shakespeare wrote only two plays, A and B . The plays are reasonably long and they are of the same length. Let X_j be the number of times that word j appears in play A , and Y_j be the number of times it appears in play B , for $1 \leq j \leq N$.

(a) Explain why it is reasonable to model X_j as being Poisson, and Y_j as being Poisson with the same parameter as X_j .

It is reasonable to model X_j and Y_j as Poisson, because this distribution is used to describe the number of "events" (such as emails received) happening at some average rate in a fixed interval or volume. The Poisson paradigm applies here: each

individual word in a play has some very small probability of being word j , and the words are weakly dependent. Here an event means using word j , the average rate is determined by how frequently Shakespeare uses that word overall. It is reasonable to assume that the average rate of occurrence of a particular word is the same for two plays by the same author, so we take λ to be the same for X_j and Y_j .

(b) Let the numbers of occurrences of the word “eyeball” (which was coined by Shakespeare) in the two plays be independent $\text{Pois}(\lambda)$ r.v.s. Show that the probability that “eyeball” is used in play B but not in play A is

$$e^{-\lambda}(\lambda - \lambda^2/2! + \lambda^3/3! - \lambda^4/4! + \dots).$$

Let X be the number of times that “eyeball” is used in play A , and Y be the number of times that it is used in play B . Since X and Y are independent $\text{Pois}(\lambda)$,

$$\begin{aligned} P(X = 0, Y > 0) &= P(X = 0)(1 - P(Y = 0)) = e^{-\lambda}(1 - e^{-\lambda}) \\ &= e^{-\lambda} \left(1 - \left(1 - \lambda + \frac{\lambda^2}{2!} - \frac{\lambda^3}{3!} + \frac{\lambda^4}{4!} - \dots \right) \right) \\ &= e^{-\lambda} \left(\lambda - \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} - \frac{\lambda^4}{4!} + \dots \right). \end{aligned}$$

(c) Now assume that λ from (b) is unknown and is itself taken to be a random variable to reflect this uncertainty. So let λ have a PDF f_0 . Let X be the number of times the word “eyeball” appears in play A and Y be the corresponding value for play B . Assume that the conditional distribution of X, Y given λ is that they are independent $\text{Pois}(\lambda)$ r.v.s. Show that the probability that “eyeball” is used in play B but not in play A is the alternating series

$$P(X = 1) - P(X = 2) + P(X = 3) - P(X = 4) + \dots$$

Hint: condition on λ and use (b).

Now λ is a random variable. Given λ , the calculation from (b) holds. By the law of total probability,

$$\begin{aligned} P(X = 0, Y > 0) &= \int_0^\infty P(X = 0, Y > 0 \mid \lambda) f_0(\lambda) d\lambda \\ &= \int_0^\infty P(X = 1 \mid \lambda) f_0(\lambda) d\lambda - \int_0^\infty P(X = 2 \mid \lambda) f_0(\lambda) d\lambda \\ &\quad + \int_0^\infty P(X = 3 \mid \lambda) f_0(\lambda) d\lambda - \int_0^\infty P(X = 4 \mid \lambda) f_0(\lambda) d\lambda + \dots \\ &= P(X = 1) - P(X = 2) + P(X = 3) - P(X = 4) + \dots \end{aligned}$$

(d) Assume that every word's numbers of occurrences in A and B are distributed as in (c), where λ may be different for different words but f_0 is fixed. Let W_j be the number of words that appear exactly j times in play A . Show that the expected number of distinct words appearing in play B but not in play A is

$$E(W_1) - E(W_2) + E(W_3) - E(W_4) + \dots$$

(This shows that $W_1 - W_2 + \dots$ is an *unbiased* prediction of the number of distinct words appearing in play B but not in play A : on average it is correct. Moreover, it can be computed just from having seen play A , without needing to know f_0 or any of the λ_j . This method can be extended in various ways to give predictions for unobserved plays based on observed plays.)

Let X_j be the number of times word j appears in play A and let W be the number of distinct words that appear in play B but not in A . Then $W = \sum_{j=1}^N I_j$, where I_j is the indicator r.v. of the event that word j appears in play B but not in play A , and N is the total number of words. By (c), for $1 \leq j \leq N$,

$$EI_j = \sum_{i=1}^{\infty} (-1)^{i+1} P(X_j = i).$$

Also, note that the number of words that appear exactly i times in play A is

$$W_i = I(X_1 = i) + I(X_2 = i) + I(X_3 = i) + \dots + I(X_N = i),$$

where $I(X_j = i)$ is the indicator of word j appearing exactly i times in play A . So

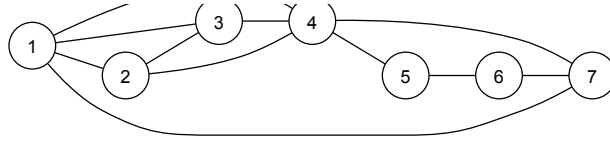
$$EW_i = \sum_{j=1}^N EI(X_j = i) = \sum_{j=1}^N P(X_j = i).$$

Then

$$\begin{aligned} EW &= \sum_{j=1}^N EI_j = \sum_{j=1}^N \sum_{i=1}^{\infty} (-1)^{i+1} P(X_j = i) \\ &= \sum_{i=1}^{\infty} (-1)^{i+1} \sum_{j=1}^N P(X_j = i) \\ &= \sum_{i=1}^{\infty} (-1)^{i+1} EW_i \\ &= EW_1 - EW_2 + EW_3 - EW_4 + \dots \end{aligned}$$

7. A *network* consists of n *nodes*, each pair of which may or may not have an *edge* joining them. For example, a social network can be modeled as a group of n nodes (representing people), where an edge between i and j means they know each other. Assume the network is undirected and does not have edges from a node to itself (for a social network, this says that if i knows j , then j knows i and that, contrary to Socrates' advice, a person does not know himself or herself). A *clique* of size k is a set of k nodes where every node has an edge to every other node (i.e., within the clique, everyone knows everyone). An *anticlique* of size k is a set of k nodes where there are no edges between them (i.e., within the anticlique, no one knows anyone else). For example, the picture below shows a network with nodes labeled $1, 2, \dots, 7$, where $\{1, 2, 3, 4\}$ is a clique of size 4, and $\{3, 5, 7\}$ is an anticlique of size 3.

(a) Form a random network with n nodes by independently flipping fair coins to decide for each pair $\{x, y\}$ whether there is an edge joining them. Find the expected number of cliques of size k (in terms of n and k).



Order the $\binom{n}{k}$ subsets of people of size k in some way (i.e., give each subset of size k a code number), and let X_i be the indicator. Since $X_1 + X_2 + \cdots + X_{\binom{n}{k}}$ is the number of cliques of size k , the expected number is

$$E(X_1 + X_2 + \cdots + X_{\binom{n}{k}}) = \binom{n}{k} E(X_1) = \binom{n}{k} P(X_1 = 1) = \frac{\binom{n}{k}}{2^{\binom{k}{2}}}.$$

(b) A *triangle* is a clique of size 3. For a random network as in (a), find the variance of the number of triangles (in terms of n).

Hint: find the covariances of the indicator random variables for each possible clique. There are $\binom{n}{3}$ such indicator r.v.s, some pairs of which are independent and other pairs of which are dependent.

Let $k = 3$ and the X_i be as in (a). Then

$$\begin{aligned} \text{Var}(X_1 + \cdots + X_{\binom{n}{3}}) &= \text{Var}(X_1) + \cdots + \text{Var}(X_{\binom{n}{3}}) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \\ &= \binom{n}{3} \text{Var}(X_1) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \end{aligned}$$

We have

$$\text{Var}(X_1) = 2^{-\binom{3}{2}}(1 - 2^{-\binom{3}{2}}) = \frac{7}{64}$$

since a $\text{Bern}(p)$ r.v. has variance $p(1 - p)$.

To compute $\text{Cov}(X_i, X_j)$ for $i < j$, consider how many people are in common for group i and group j . If the number of people in common is 0 or 1 (as shown in the upper and lower left cases in Figure 1, respectively), then the $\text{Cov}(X_i, X_j) = 0$ since the coin flips used to determine whether Group i is a clique are independent of those used for Group j . If there are 2 people in common (as shown in the lower right case of Figure 1), then

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j) = \frac{1}{2^5} - \left(\frac{1}{2^3}\right)^2 = \frac{1}{64},$$

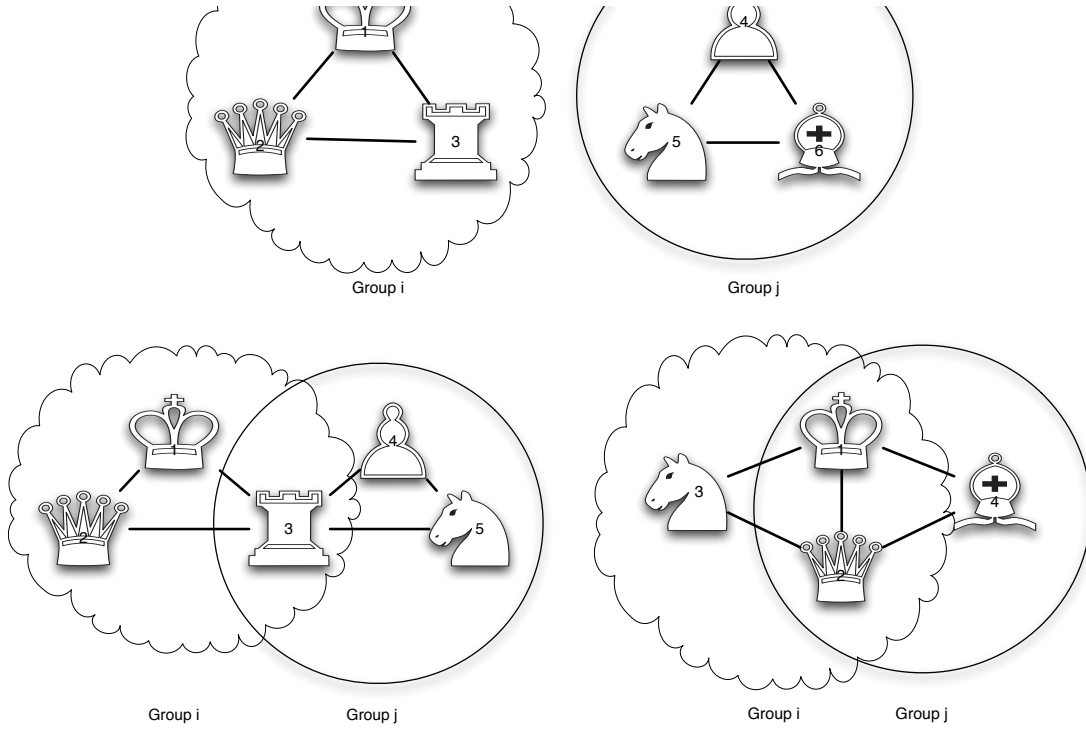


Figure 1: Two groups with 0 (upper), 1 (lower left), 2 (lower right) people in common.

since 5 distinct pairs of people must know each other to make $X_i X_j$ equal to 1.

There are $\binom{n}{4} \binom{4}{2} = 6 \binom{n}{4}$ pairs of groups $\{i, j\}$ ($i \neq j$) with 1 pair of people in common (choose 4 people out of the n , then choose which 2 of the 4 are the overlap of the groups). The remaining pairs of groups have covariance 0. Thus, the variance of the number of cliques is

$$\frac{7}{64} \binom{n}{3} + 2 \cdot 6 \binom{n}{4} \cdot \frac{1}{64} = \frac{7}{64} \binom{n}{3} + \frac{3}{16} \binom{n}{4}.$$

(c) Suppose that $\binom{n}{k} < 2^{\binom{k}{2}-1}$. Show that there is a network with n nodes containing no cliques of size k or anticliques of size k .

Hint: explain why it is enough to show that for a random network with n nodes, the probability of the desired property is positive; then consider the complement.

We will prove the existence of a network with the desired property by showing that the probability is positive that a random network has the property is positive. Form a random network as in (a), and let A_i be the event that the i th group of k people (in any fixed ordering) is neither a clique nor an anticlique. We have

$$P\left(\bigcup_{i=1}^{\binom{n}{k}} A_i^c\right) \leq \sum_{i=1}^{\binom{n}{k}} P(A_i^c) = \binom{n}{k} 2^{-\binom{k}{2}+1} < 1,$$

which shows that

$$P\left(\bigcap_{i=1}^{\binom{n}{k}} A_i\right) = 1 - P\left(\bigcup_{i=1}^{\binom{n}{k}} A_i^c\right) > 0,$$

as desired. Alternatively, let C be the number of cliques of size k and A be the number of anticliques of size k , and write $C + A = T$. Then

$$E(T) = E(C) + E(A) = \binom{n}{k} 2^{-\binom{k}{2}+1} < 1$$

by the method of Part (a). So $P(T = 0) > 0$, since $P(T \geq 1) = 1$ would imply $E(T) \geq 1$. This again shows that there must be a network with the desired property.